Exact canonicalization of configured molecular stereographs

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Abstract

Background: Connectivity-only molecular graphs require additional constraints to represent configuration. Relative ligand frames and stereo-aware canonicalization are established approaches; a uniform reduction needs explicit guarantees about transport, global symmetry, represented supports and incomplete information.

Methods: We define admissible configured molecular stereographs with typed atom, bond, axis, path, plane and coupled-frame supports, then encode their configuration orbits in a colored auxiliary graph. Canonical labeling gives a versioned certificate compared by full equality. Declared mirror actions supply molecular relations. Attributed constitutional identity, finite resonance-family identity, perception and downstream naming have separate contracts; unresolved strict inputs and bounded computations refuse a definitive result.
Results: Independent direct assignment-orbit checks agree with the complete production partitions on 63 fixtures and 2439 assignments. Every stored transport witness replays, mirror and renumbering checks agree, and all 7 deliberate defects are rejected. Public fixtures are conditional on their supplied carrier inventories. Independent geometric annotation certifies 38 of 40 newly sourced coordination examples; every certified source/mirror and renumbering comparison agrees. All 48 sourced constitutional negatives and 60 frozen, source-assisted structural cases pass their declared checks. Controlled registration recovers the independent groups but shows no speed advantage for certificates.
Conclusions: The construction makes local and simultaneous configurations comparable under a declared identity convention. Independent partition evidence supports the configured-graph kernel; chemical perception, naming coverage and physical configurational stability require their own validation.
Scientific Contribution: We give a uniform faithful reduction for the declared admissible support catalogue, with explicit transport, mirror and information contracts. An independent assignment-partition oracle validates both separation and identification, including simultaneous configurations and coupled frames.

Keywords: chemical graph, stereochemistry, graph canonicalization, permutation group, graph isomorphism, molecular chirality, CIP nomenclature

1 Introduction

Canonical molecular identity underlies registration, database deduplication, interchange, and reproducible modeling. Molecular graphs and canonical line notations provide the usual foundation [1–4], but an ordinary attributed graph does not distinguish stereoisomers: two enantiomers can have identical atoms, bonds, bond orders, and charges. Appending CIP R/S or E/Z does not make those names primary identity, because they are derived from constitution-dependent priority rules [5, 6]. A binary parity is also insufficient for coordination geometries with more than two configurations.

Ordered ligand frames modulo geometry-preserving permutation groups give coordinate-free relative stereo [7, 8]. For tetrahedra, the even subgroup A_4 and its odd coset represent the two configurations. Existing toolkits operationalize perception, serialization, naming and stereo-aware canonicalization [3, 9–11]. These are prior foundations.

The research gap appears when local states must become one molecular identity. A local orbit can accept or reject a *proposed* atom mapping, but it does not choose a canonical global mapping, resolve several symmetry-coupled loci, or yield a complete certificate independent of atom and descriptor order. General graph canonicalization resolves global symmetry by refinement and individualization [12, 13], and molecular atom-ordering algorithms provide canonical constitutional graphs [2, 4]. Neither result alone proves that geometry-specific local stereo orbits are preserved in the canonical object for every support type. We establish a uniform faithful reduction from the *configured stereograph* for the declared support catalogue to ordinary colored-graph isomorphism.

We provide that link. Each local orbit is expanded into explicit relations between tuples and positions attached to typed ligand ports, and all loci are joined to the same LLG constitution. Exact colored-graph canonicalization then resolves local and global symmetry simultaneously. Writing Φ_χ for this expansion, $\cong$ for colored-graph isomorphism, $\cong_{\mathrm{st}}$ for stereo-isomorphism, and C^{χ}_{SG} for the canonical certificate, we prove

$$\mathcal{X} \cong_{\mathrm{st}} \mathcal{Y} \iff \Phi_\chi(\mathcal{X}) \cong_{\mathrm{col}} \Phi_\chi(\mathcal{Y}) \iff C^{\chi}_{\mathrm{SG}}(\mathcal{X}) = C^{\chi}_{\mathrm{SG}}(\mathcal{Y}). \tag{1}$$

The first equivalence establishes faithfulness and the second canonical-form exactness. Full serialization equality therefore decides molecular identity independently of atom numbering and local representatives.

The pipeline perceives typed carriers, attaches authorized configuration, canonicalizes the complete stereograph, and derives mirror relations and optional CIP names. Incomplete and unsupported information remains explicit. The contributions are:

1. the faithful encoding Φ_χ for ten configured families and its exact certificate C^{χ}_{SG}.
2. global identity and mirror relations from full certificate equality, with replayable automorphism maps.
3. a configuration-neutral input contract with typed atom, bond, axis, path, and plane supports, owner-scoped virtual hydrogen and lone-pair references, and explicit incomplete and unsupported outcomes.
4. double-coset local relations $\Gamma_\tau \pi \Gamma_\tau$, replayable transport witnesses, and CIP projections with deciding evidence.
5. a unified, fail-closed algorithm and soundness theorem connecting configuration-neutral perception, authorized configuration attachment, exact canonicalization, and molecular chirality.

Reaction transport lies outside the scope of this molecular identity study. The methods establish the representation, faithful reduction and mirror classification; task-specific experiments follow. Detailed enumerations, computational contracts and executable obligations appear in the supporting information (Additional file 1).

2 Methods

2.1 Use of AI assistance

OpenAI Codex assisted the revision through code and mathematical review, implementation, test and analysis scripts, interpretation of computational outputs, figure preparation, and manuscript drafting. The machine-readable reports distinguish automated mathematical annotation from independent human review. Final author review of the revised content remains required before submission.

2.2 Preliminaries and notation

2.2.1 Lewis-labeled constitutional substrate

All graphs are finite. Following the Lewis-labeled graph formalism introduced for executable electron transfers in the companion graph-rewriting layer [14], the constitutional substrate is a *Lewis-labeled graph*

$$G = (V(G), E(G), a_G, b_G), \qquad a_G(v) = (\mathrm{el}_v, \ell_v, r_v), \qquad b_G(e) = (s_e, p_e), \tag{2}$$

where el_v is the element symbol, $\ell_v, r_v \in \mathbb{N}_0$ are the numbers of lone pairs and radical electrons, and $s_e, p_e \in \mathbb{N}_0$ are the σ- and π-bond orders. The optional electron-resource bookkeeping is specified in Appendix B.1. Separating σ and π resources retains the distinction between a cumulated double-bond path and a single-bond chain. Identity is determined by the attributes selected below.

An *identity profile* χ selects invariant colors from a fully declared attributed substrate: the LLG data together with isotope, hydrogen, radical, and aromatic attributes when those are part of the request. Formally, a drawing-level profile assigns vertex and edge colors by a rule that commutes with relabeling,

$$\chi_{fG}(fx) = \chi_G(x) \qquad \text{for every isomorphism } f \text{ and every } x \in V \cup E. \tag{3}$$

We write G_χ for the resulting colored constitution, $\mathrm{Iso}_\chi(G, H)$ for the adjacency- and color-preserving bijections, and $\mathrm{Aut}_\chi(G) = \mathrm{Iso}_\chi(G, G)$. Write $\chi_1 \preceq \chi_2$ when every χ_1 color is obtained from its χ_2 color by a fixed relabeling-invariant map. Only for such an ordered pair does refinement give $\mathrm{Iso}_{\chi_2}(G, H) \subseteq \mathrm{Iso}_{\chi_1}(G, H)$. Bare LLG colors are not automatically finer than an isotope-sensitive profile. The profile is part of the request: strict Lewis identity, chemical resonance-family identity, and connectivity-only comparison are different, versioned questions rather than silent normalizations. Eq. (3) is

what makes every downstream invariance statement in this paper meaningful; a profile computed from input atom order would invalidate all of them.

Resonance-family identity is a separate extension of drawing-level colors. For an explicitly declared finite family $\mathcal{R}(\mathcal{X})$, its key is the versioned sorted set of full drawing certificates

$$C_{\mathcal{R}}(\mathcal{X}) = \mathrm{Serialize}(\{C_{\mathrm{SG}}(\mathcal{Z}) : \mathcal{Z} \in \mathcal{R}(\mathcal{X})\}) .$$

It identifies the family of configured drawing-isomorphism classes, rather than a selected drawing or merely an intersection with another family. The graphification theorem below applies to each drawing. Numbering invariance of the family extension additionally requires an equivariant family constructor. The implementation declares RDKit resonance enumeration with charge separation and all Kekulé forms, closed under the specified terminal phosphate charge/bond-order exchange. It preserves isotope and atom identity, and refuses when enumeration reaches its safety cap. No claim of exhaustive enumeration of every chemically conceivable resonance state is made. A family member's atom order and automorphisms remain member witnesses; they are not a unique ordering or the full automorphism group of the family.

A structural morphism preserves adjacency and all colors selected by χ. An isomorphism is bijective and also reflects adjacency. We use $\varphi : G \to H$ for a general graph map; the shorter letters f, g in transport formulas denote maps of the same kind. Ordinary function composition is right-to-left. Vertex identifiers are separate from the chemical labels a_G, b_G, and renumbering does not change those labels. When the graph is fixed, V, E, a, b abbreviate these graph-indexed objects. Table 1 collects the shared graph notation and the additional stereochemical objects.

Table 1 Graph and stereochemical notation.

Symbol	Meaning
$V(G), E(G), a_G, b_G$	Vertices, edges, atom labels and bond labels
ℓ_v, r_v, s_e, p_e	Lone pairs, radicals, and σ/π bond resources
$\varphi : G \to H$, $G \cong H$	A graph map and existence of a graph isomorphism
$\mathrm{Aut}_\chi(G)$	Constitution automorphisms under profile χ
$\mathcal{X} = (G, \mathcal{R})$	Constitution with its typed configuration registry
$\Gamma_{\tau_s}, [u_s]_{\Gamma_{\tau_s}}$	Registered geometry group and configuration orbit
$\Phi_\chi(\mathcal{X})$	Colored auxiliary graph encoding the configured object
$\mathrm{Can}(K), \mathrm{can}(K)$	Canonical representative graph and its lossless encoding
$C^\chi_{\mathrm{SG}}(\mathcal{X})$, $\mu\mathcal{X}$	Full configured certificate and declared mirror

2.2.2 Configured stereographs and permutation actions

A configured molecular stereograph is

$$\mathcal{X} = (G, \mathcal{R}),$$

where G is a finite undirected simple attributed constitution and $\mathcal{R}$ is a finite registry of typed stereo loci. Each registered carrier key has at most one descriptor. Paths have distinct material atoms and continuous selected bonds; plane cycles have distinct atoms and a valid pilot; framework supports are connected and pairwise disjoint, with distinct frame centers. Configuration is fixed for the binary identity/mirror domain; unknown and unsupported input belongs to an explicit information boundary. A finite-frame locus is

$$s = (\tau_s, P_s, \Gamma_{\tau_s}, [u_s]_{\Gamma_{\tau_s}}),$$

with geometry type τ_s, ordered material or virtual ligand ports P_s, a geometry-preserving group $\Gamma_{\tau_s} \leq S_{k_s}$, and one configuration orbit. Path- and plane-supported families use the analogous canonical reversal/rotation forms.

Here S_k denotes the symmetric group on k positions. Concrete subgroups such as A_4, D_4, D_3, V_4, and proper O refer to registered permutation actions, not merely to abstract isomorphism types. Owner-scoped virtual references `@H:`x and `@LP:`x allow

an implicit hydrogen or lone pair to occupy a ligand position without inventing a phantom atom or allowing the resource to migrate between owners.

A *frame* of length k over a reference set R is an injective tuple $u = (u_1, \ldots, u_k) \in R^k$; equivalently, a bijection from the slot set $\{1, \ldots, k\}$ onto its image $\operatorname{im} u \subseteq R$. Injectivity is a typing requirement: a slot is a coordination position, and no reference may occupy two positions of the same locus. For a permutation $p \in S_k$ and a frame u we use

$$(p \cdot u)_i = u_{p(i)}, \qquad p \text{ then } q = p \circ q. \tag{4}$$

The slot convention makes sequential tuple reorderings satisfy $(p\text{then}q)\cdot u = q\cdot(p\cdot u)$, although their slot images compose as $p \circ q$. Entrywise reference relabeling commutes with these reorderings. Injective frames have unique alignments, so a geometry group acts freely and every local orbit has its group order. For tetrahedral geometry, the two A_4 orbits correspond to even and odd alignments. Appendix A proves these action, alignment and parity laws, including the convention needed for noncommuting transports. Figure 1 illustrates the two alignments in one fixed geometric frame.

This representation keeps three concepts separate. A *support* states where an element lives. A *configuration* is the orbit or canonical path or plane state. A *derived label* names that state relative to a priority convention. Derived labels depend on constitution and configuration. They do not enter stored identity or canonicalization. Candidate-carrier evidence is not a configuration. Unknown orientation is not a racemic mixture. Descriptor absence is not a query wildcard.

Figure 2 shows why that separation matters. Connectivity cannot distinguish either the tetrahedral pair or the planar-bond pair. An allene requires an axis. Helicity requires a path. Planar chirality requires an oriented plane. Topological or mechanical stereo may require a whole component.

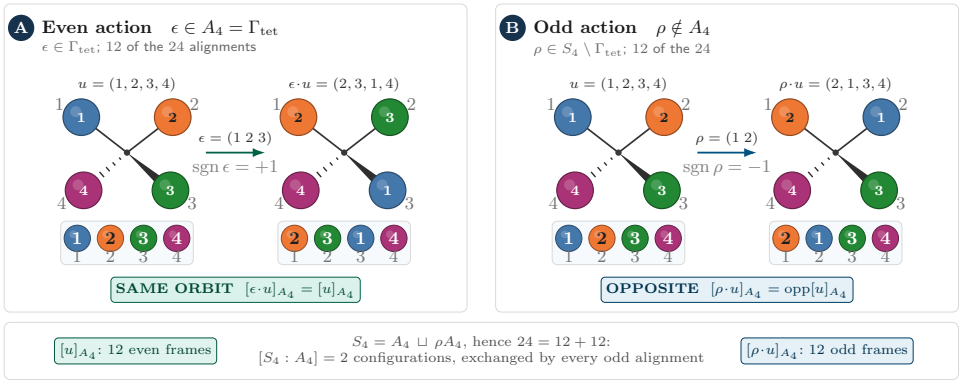


Fig. 1 Even and odd alignments of one ordered four-ligand frame. Colored numbered balls identify ligand references, the gray digit beside each position identifies the frame slot, and the strip below each center repeats the same state as the ordered tuple read in slot order. All four centers share one fixed geometric frame. It contains two in-plane bonds, a bold wedge at slot 3 toward the viewer, and a hashed bond at slot 4 away from it. Only the assignment of ligands to slots changes. (A) The even 3-cycle $\epsilon = (1\ 2\ 3) \in A_4 = \Gamma_{\text{tet}}$ sends $u = (1, 2, 3, 4)$ to $(2, 3, 1, 4)$. Green marks a configuration-preserving action, a rigid rotation of the same local state. (B) The odd transposition $\rho = (1\ 2) \notin A_4$ sends u to $(2, 1, 3, 4)$. Blue marks the opposite orbit and uses the same verdict colors as Figure S1. The ledger states the coset decomposition $S_4 = A_4 \sqcup \rho A_4$ behind Proposition S3: four distinct references admit 24 ordered frames that split $12 + 12$ into two mirror-image configuration orbits.

2.3 Related frameworks

The ordered-frame group action follows the `MØD` stereochemistry extension [7], which represents partial information through shape morphisms and treats tetrahedral examples in detail. Molassembler already provides stereochemistry-aware molecular comparison and canonicalization: its modular atom colors incorporate shapes and stereopermutations, and `nauty` canonical labeling is applied to the graph and its stereopermutators [11]. Thus, canonical molecular stereo identity and permutation-based coordination models are established prior contributions. MACE also identifies coordination configurations and mirrors through sets of canonical simplified molecular-input line-entry system (SMILES) strings generated under the proper rotations of square-planar and octahedral donor labels [15]. `StereoMolGraph` represents relative configurations and performs whole-graph comparison with a stereochemistry-aware VF2++-style algorithm [8, 16]. Papusha and Leonhard also describe resonance-invariant circular stereo refinement and fingerprints in the SACA preprint [17].

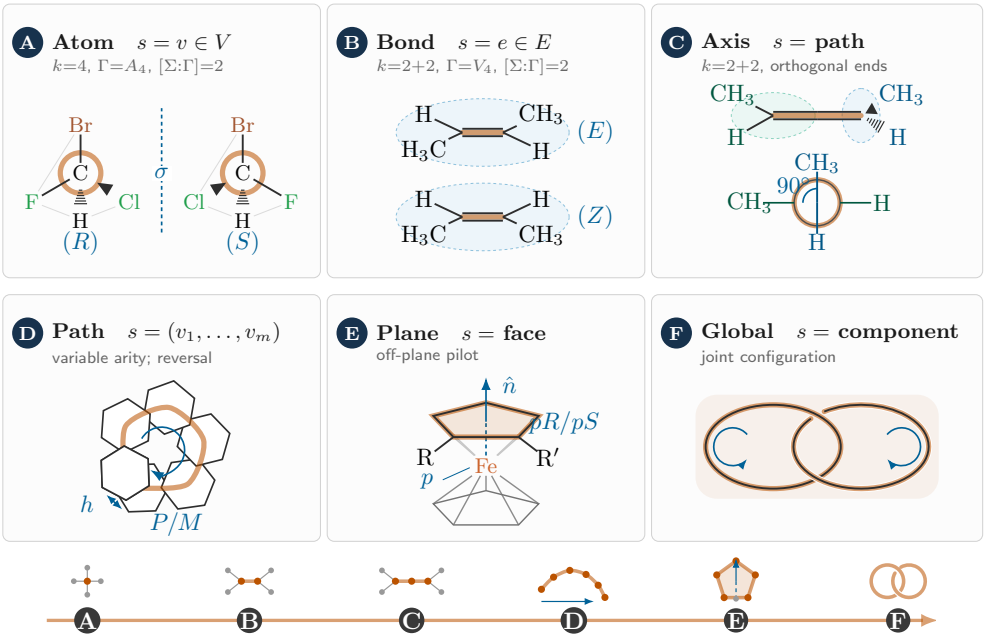


Fig. 2 Typed supports for configured stereochemical elements. The support of a stereo element is a subgraph of prescribed shape, and the six families differ in that shape rather than in their algebra. Each header states the support type together with the arity and geometry group where the family uses a finite frame. Vermilion marks the carrier, blue marks the orientation datum, and green distinguishes the second terminal plane in panel C. (A) An atom support is one vertex; the two tetrahedral states share a single constitution and are exchanged by the mirror σ. (B) A bond support is one edge carrying two terminal pairs. (C) An axis support is a cumulated path whose termini lie in orthogonal planes, shown here in projection along the axis. (D) A path support is an ordered vertex sequence taken modulo canonical reversal; helicity is the sense of the rise h. (E) A plane support is an oriented face whose off-plane pilot p fixes the sense of the normal $\hat{n}$. (F) A global support couples frames over a whole component; its joint configuration is retained explicitly. The footer ramp orders the six carriers by spatial extent and defines no inheritance between them. Molecular chirality, including a *meso* verdict, is a downstream whole-stereograph classification rather than a seventh carrier family.

Fingerprint indexing and similarity remain distinct from equality of a full canonical certificate; exact `StereoMolGraph` isomorphism is a separate operation. The `StereoMolGraph` connectivity-focused convention differs from a fully attributed Lewis-state identity. `RDKit` supplies canonical stereo representations, including atropisomer canonicalization, separately from CIP assignment [9, 18]. Release-specific representation probes in RDKit 2026.03.1 retain the distinct square-planar, trigonal-bipyramidal and octahedral states of the documented distinct-ligand examples. Their canonical SMILES agree across every tested atom permutation and parse/write round trip. These finite checks concern represented configurations; they do not assess chemical

perception, coordinate interpretation or naming, and do not prove canonicalization for every possible input. The probe source, fixtures and report accompany the revision archive. `CGRtools` addresses stereo in reaction-graph processing [10]. This layer builds on the `SynKit` platform [19].

Table 2 Documented approaches closest to the configured-identity task. Entries describe the cited representation or algorithm, rather than asserting that unlisted capabilities are absent. Different information conventions require matched profiles before empirical comparison.

Approach	Representation and scope	Identity or transport operation
`MØD` [7]	Ordered neighbors, shape groups and partial configurations	Configuration-compatible graph morphisms and rewriting
Molassembler [11]	Atom/bond stereopermutators; organic and coordination shapes	Modular colored isomorphism and canonical labeling
MACE [15]	Donor-labeled square-planar and octahedral configurations	Rotation-related canonical-SMILES sets; duplicate and mirror screening
`RDKit` [18]	Stereo tags; separate format, family and naming support	Canonical stereo representations and CIP assignment
`StereoMolGraph` [8, 16]	Relative stereo on connectivity-focused graphs	Extended whole-graph isomorphism; fingerprint indexing
This work	Declared profiles; finite frames, paths, planes and coupled frameworks	Faithful auxiliary encoding, full certificates and replayable transport

2.4 Problem statement

Existing frameworks already lift local configurations into molecular comparison and, in some cases, canonicalization. Our contribution is a uniform faithful reduction for a declared catalogue of support types, with explicit information, profile and transport contracts. Local permutation invariance alone does not establish this reduction: both preservation and reflection of global stereo-isomorphism must be proved, including coupled supports and molecular symmetry. For that declared domain, we require a

computable map C^{χ}_{SG} on configured stereographs such that

$$C^{\chi}_{\mathrm{SG}}(\mathcal{X}) = C^{\chi}_{\mathrm{SG}}(\mathcal{Y}) \iff \mathcal{X} \cong_{\mathrm{st}} \mathcal{Y}, \tag{5}$$

$$C^{\chi}_{\mathrm{SG}}(f\mathcal{X}) = C^{\chi}_{\mathrm{SG}}(\mathcal{X}) \quad \text{for every relabeling } f, \tag{6}$$

where $\cong_{\mathrm{st}}$ is the stereo-isomorphism of Definition 1 under the declared profile χ. The mathematical map is total on the declared admissible configured domain. Its bounded implementation returns an exact value only after a completed search and otherwise refuses; belonging to an audited corpus does not guarantee completion within a resource budget. Condition (5) is completeness and soundness of the certificate, and Eq. (6) excludes any dependence on input numbering, on the choice of local representative, or on derived labels. Theorem 5 constructs such a certificate by reducing $\cong_{\mathrm{st}}$ faithfully to colored-graph isomorphism.

Figure 3 states the information flow used here. Perception may identify a carrier, but only authorized evidence may attach a configuration. Local states are then graphified together and canonicalized exactly. Molecular classification and derived naming consume that identity. Neither operation feeds information backward. The central arrow is the configured stereograph canonicalization developed in Section 2.6.

2.5 Stereochemical supports and configurations

2.5.1 Configured support families

A stereo support is an immutable carrier together with its dependency set and relabeling action. Table 3 summarizes the configured families retained in the main text. Appendix G through Appendix H.2 gives the validation invariants and serialization details.

The support headings classify carrier semantics rather than set inclusion. A bond is a path of length one as an underlying graph object. Bond support and path support

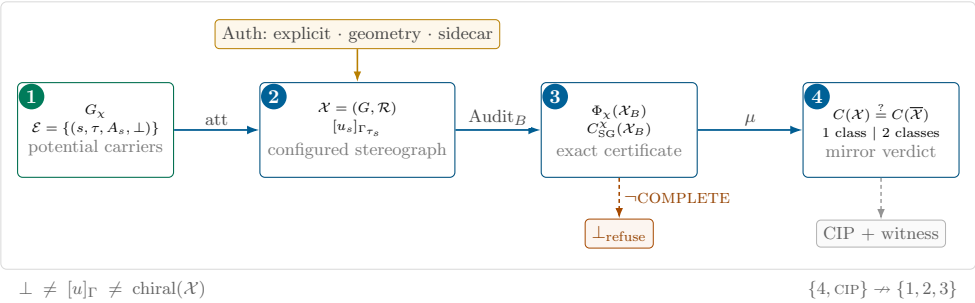


Fig. 3 One-way, fail-closed information flow, written as typed objects and the maps between them. Stage 1 is configuration-neutral perception and returns potential carriers with configuration $\perp$. Only authorized evidence Auth may apply the attachment map att of Definition 2, which produces the configured stereograph $\mathcal{X}$. The declared support boundary B is audited before the expansion Φ_χ and the exact certificate C^{χ}_{SG}; a failed audit refuses instead of returning a partial identity. The mirror map μ compares the certificate of $\mathcal{X}$ with that of its mirror image $\overline{\mathcal{X}}$ and returns one or two classes. Derived CIP naming is an optional consumer. The footer states the two structural invariants: absent, configured, and molecularly chiral are three distinct predicates, and no downstream stage feeds information back into stages 1–3.

Table 3 Configured stereochemical families grouped by support. For a finite-frame family, k_τ is the arity, Σ_τ the frame symmetry group of admissible slot rearrangements, and Γ_τ the geometry-preserving group; the state count is then the index $[\Sigma_\tau : \Gamma_\tau]$ of Proposition S6. Path and plane families use canonical reversal or rotation forms instead of a finite frame quotient. The separately versioned global support is not counted as an additional local family.

Family	k_τ	$\lvert\Sigma_\tau\rvert$	Γ_τ (order)	States	Example
Atom support					
Tetrahedral	4	24	A_4 (12)	2	C, P, S, N
Square planar	4	24	D_4 (8)	3	Pt(II) complex
Trigonal bipyramidal	5	120	D_3 (6)	20	P, 5-coord.
Octahedral	6	720	proper O (24)	30	metal, 6-coord.
Bond support					
Planar bond	$2+2$	8	V_4 (4)	2	alkene, imine
Axis support					
Atrop axis	$2+2$	8	V_4 (4)	2	hindered biaryl
Cumulene axis	$2+2$	8	V_4 (4)	2	allene, cumulene
Path support					
Extended cis/trans	$2+2$	—	canonical reversal	2	odd cumulene
Helical	variable	—	reversal, rotation	two senses	helicene
Plane support					
Planar chirality	variable	—	oriented cycle	two senses	planar-chiral ring

nevertheless remain distinct types. Bond support uses terminal pair frame algebra, whereas path support uses ordered traversal and reversal semantics. The progression in Figure 2 records spatial extent and does not define an inheritance hierarchy.

The carrier is assessed before assigning its focal configuration. The constitution-only constructor $K_0(G, s)$ excludes all supplied configuration, CIP properties, atom-map labels as identity, and coordinates. The retained protocol instead uses $K_{\mathcal{F}_{-s}}(G, s)$, where neighboring oriented frames are supplied and the focal frame is omitted. This second protocol is *focal-configuration-neutral*. Its stabilizer is $\mathrm{Aut}(G, \mathcal{F}_{-s})_s$; constitution-only assessment uses $\mathrm{Aut}(G)_s$. Each constructor returns frame-orbit or tie evidence when symmetry prevents an equivariant concrete ordering. For an attributed-graph isomorphism $f : G \to H$ transporting the neighboring frames,

$$K_{f\mathcal{F}_{-s}}(fG, fs) = fK_{\mathcal{F}_{-s}}(G, s).$$

Conjugation by f carries the corresponding constrained stabilizers to one another. The evidence is therefore numbering invariant without requiring a unique equivariant tuple. Setting $\mathcal{F}_{-s} = \varnothing$ recovers the constitution-only statement.

The support table specifies representation scope. The supplementary chemical carrier gallery in Appendix C illustrates that scope; it does not independently validate handedness, persistence or automatic perception for every depicted family.

2.5.2 Input information and required configuration

A potential carrier is evidence about a support, not an assigned configuration. Constitution-only assessment excludes all supplied configurations. A separate focal-configuration-neutral protocol may retain oriented neighboring frames, while excluding the focal frame, CIP properties and input indices. Its evidence must transport with those neighboring constraints. Symmetric inputs may return a frame orbit or a tie; a unique equivariant ordered tuple is not required.

Attachment accepts configuration only from a typed explicit encoding, independently validated geometry or declared sidecar. Support, arity, ports, virtual-resource ownership and existing registry information must agree. Conflicting evidence refuses rather than changing a locus silently. The formal attachment contract and its equivariance proof appear in Appendix A.3; the perception predicate and conditional equivariance statement appear in Appendix A.2.

The requested completeness boundary is part of the input. A source-declared boundary assesses exactly the represented configurations; additional undeclared potential loci are outside that claim. A strict boundary requires every supported locus selected by the active family policy to be configured. A required unknown or unsupported locus prevents a binary mirror verdict. Thus partially configured 2,3-dichlorobutane remains incomplete, because its possible completions include both a meso arrangement and chiral arrangements. Neither an unequal partial mirror certificate nor an atom-count heuristic proves that all admissible completions are chiral. A separate completion-population operation addresses that quantified question. Figure 4 displays the two completions of one fixed partial input and the distinct questions answered by each boundary.

2.5.3 Finite-frame configuration orbits

For a geometry τ, Σ_τ comprises admissible slot rearrangements and $\Gamma_\tau \leq \Sigma_\tau$ preserves configuration. Atom geometries admit all ligand permutations. Bond and terminal-axis frames admit within-end exchanges and endpoint exchange with their anchors. Distinct references give $[\Sigma_\tau : \Gamma_\tau]$ local configurations before constitutional symmetry identifies any of them. The counts in Table 3 concern distinct slots, not necessarily distinguishable chemical ligands.

An index-two subgroup supplies one unique other state. For larger indices, the index alone selects no opposite; additional geometric structure can still supply a

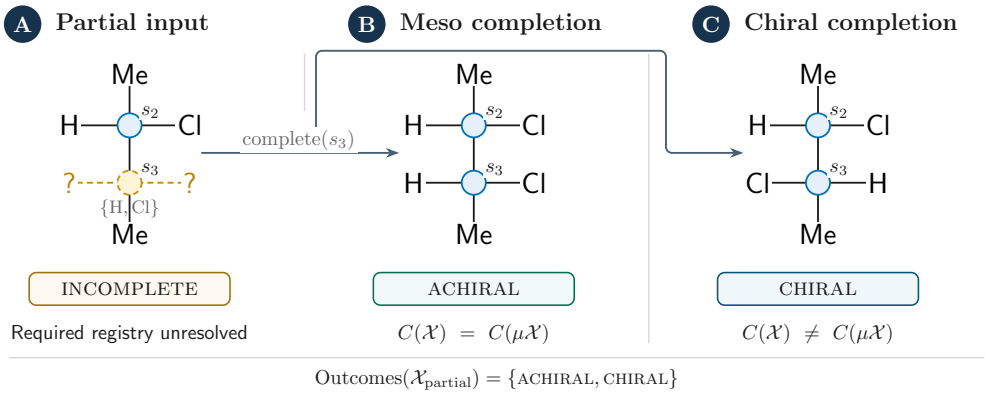


Fig. 4 Incomplete configuration permits both achiral and chiral completions. (A) A partially configured 2,3-dichlorobutane fixes the upper center and leaves the lower required center unresolved. (B,C) Completing the lower center yields a meso configuration or a chiral configuration. The drawings use Fischer projections: horizontal bonds point towards the viewer and vertical bonds away; dashed lower arms in (A) indicate missing configuration, not a geometric assignment. The strict input therefore returns incomplete. A source-declared assessment concerns only its represented registry; a completion-population assessment quantifies over the admitted completions. These are different questions under the same declared identity profile.

mirror involution. TBP and octahedral mirrors use their registered improper extensions. Square-planar and planar-bond reflection preserves the local orbit. In particular, the binary change $E \leftrightarrow Z$ is not geometric reflection. Appendix A.4 proves the configuration count and binary-opposite statements and illustrates admitted frame changes.

2.5.4 Extended axis, path, and plane configurations

Atrop and cumulene axes use the same four-reference V_4 frame algebra but different support and admissibility conditions. A cumulene-axis descriptor retains the complete cumulated path and is admitted for an even number of double bonds. An ordinary alkene is bond-supported. Helical paths have variable arity, so identity is the minimum over path reversal and, for cyclic paths, rotation. Planar chirality analogously minimizes an oriented cycle while coupling reversal to parity inversion. These families assign no configuration or stability from connectivity alone. The chemical cases, canonical forms, and reversal laws are detailed in Appendix H.2 through Appendix I.3.

2.5.5 Local relations and transport evidence

A transport witness replays a concrete reference alignment. Replacing either representative within its geometry orbit changes that witness by left and right multiplication in Γ_τ, leaving the double coset $\Gamma_\tau \pi \Gamma_\tau$ invariant. Witnesses compose with the declared `then` convention; double cosets classify endpoint relations and generally do not have a single-valued composition. Stored fixed, unknown, unspecified and absent information remains distinct. Query wildcards are an explicit matching policy, not stored equality. Detailed relation and morphism contracts are in Appendices A.5 and J.

2.6 Exact graph canonicalization

The central task is to turn $\mathcal{X} = (G, \mathcal{R})$ into one exact, numbering-independent identity. Local orbit canonical forms can verify a proposed structural isomorphism, but they do not resolve global symmetry or choose a canonical atom order. We instead reduce the complete configured stereograph to one colored auxiliary graph and delegate the local–global symmetry interaction to a complete graph canonizer.

2.6.1 One global map must satisfy every local constraint

A constitutional bijection $f : G \to H$ transports all configured supports simultaneously. If u_s is a source frame and $v_{f(s)}$ an image frame, there is a unique admissible alignment $\rho_s(f)$ satisfying $f(u_s) = \rho_s(f) \cdot v_{f(s)}$. Configuration is preserved precisely when $\rho_s(f) \in \Gamma_{\tau_s}$. This membership is independent of the chosen orbit representatives.

Several local assignments cannot be compared independently: one atom map must satisfy every membership condition, including when constitutional symmetry exchanges entire loci. The induced alignments obey the cocycle law proved in Appendix A.6, so valid transports are closed under composition and inversion. The auxiliary expansion below encodes these simultaneous requirements in one colored

graph. An automorphism of that graph therefore resolves whole-molecule symmetry, including meso exchange, rather than selecting a local ordering first.

2.6.2 Raw, formal, and global quotients

Raw encodings choose an ordered frame at each locus. Formal assignments choose a local preserving-group orbit, removing equivalent descriptions of that state. For a fixed typed substrate, write

$$\mathcal{R} = \prod_s \mathcal{R}_s, \qquad \mathcal{F} = \prod_s (\mathcal{R}_s/\Gamma_{\tau_s}), \qquad \mathcal{C} = \mathcal{F}/\operatorname{Aut}_\chi(G, S, \tau).$$

The second quotient applies one constitutional automorphism simultaneously to all loci. It may exchange entire carriers and has fixed points, including meso assignments; dividing the number of formal assignments by the group order is therefore generally incorrect. The transport action and orbit-counting formula are proved in Appendix B.2. Corollary S13 proves that raw and formal enumeration induce the same global partition. Raw multiplicity adds representations, rather than chemical configurations. Figure 5 shows the two quotients and the separately scoped validation populations.

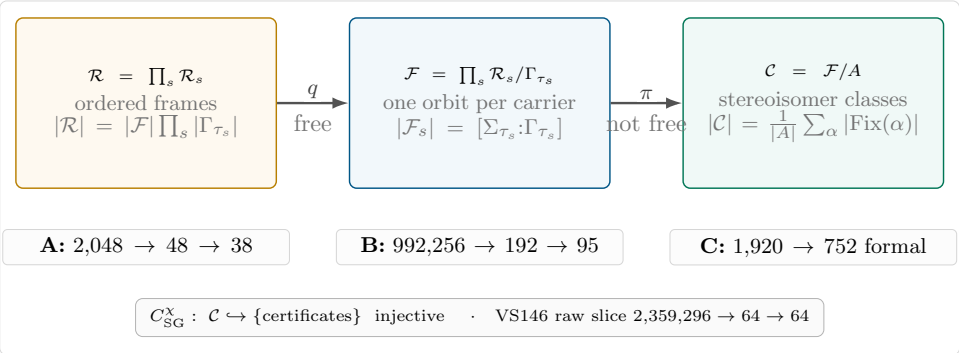


Fig. 5 The two-stage quotient used in theory and validation. Budgets A and B exhaust the displayed raw products. Formal C exhausts all nine C cases; the raw C result is deliberately labelled as the complete VS146 slice rather than as the infeasible full C product. Every reported raw encoding maps to its formal tuple and then to the same exact global class.

2.6.3 Stereo-isomorphism

Definition 1 (Stereo-isomorphism)**.** Let $\mathcal{X} = (G, \mathcal{R}_\mathcal{X})$ and $\mathcal{Y} = (H, \mathcal{R}_\mathcal{Y})$ be configured stereographs under a declared profile χ. A bijection f is a *stereo-isomorphism* when

(S1) $f \in \mathrm{Iso}_\chi(G, H)$;

(S2) f induces a bijection $\mathcal{R}_\mathcal{X} \to \mathcal{R}_\mathcal{Y}$, $s \mapsto f(s)$, with $\tau_{f(s)} = \tau_s$, and carries the ordered port incidences P_s onto $P_{f(s)}$, respecting owner scope for virtual references without fixing an arbitrary tuple ordering;

(S3) $\rho_s(f) \in \Gamma_{\tau_s}$ for every finite-frame locus s, where $\rho_s(f)$ is the induced alignment of Lemma S9;

(S4) f carries the canonical form of every path- and plane-supported locus to that of its image, that is, it respects the declared reversal, rotation, and orientation identities. For a coupled framework, it transports the support and all frame centers and preserves their jointly configured frame orbits.

We write $\mathcal{X} \cong_{\mathrm{st}} \mathcal{Y}$ when such an f exists, and $\mathrm{Aut}_{\mathrm{st}}(\mathcal{X})$ for the set of stereo-isomorphisms $\mathcal{X} \to \mathcal{X}$.

Lemma 1 ($\cong_{\mathrm{st}}$ is an equivalence relation)**.** *Stereo-isomorphisms contain the identity and are closed under composition and inverses. Hence* $\cong_{\mathrm{st}}$ *is an equivalence relation on configured stereographs and* $\mathrm{Aut}_{\mathrm{st}}(\mathcal{X})$ *is a subgroup of* $\mathrm{Aut}_\chi(G)$.

Proof. Conditions (S1), (S2), and (S4) are preserved by composition and inversion because each is a conjunction of type and incidence constraints on bijections. For (S3), Lemma S9 gives $\rho_s(\mathrm{id}) = \mathrm{id} \in \Gamma_{\tau_s}$, $\rho_s(g \circ f) = \rho_{f(s)}(g)$ then $\rho_s(f)$, and $\rho_{f(s)}(f^{-1}) = \rho_s(f)^{-1}$. Since each Γ_{τ_s} is a subgroup of Σ_{τ_s}, it contains the identity and is closed under composition and inverses, so (S3) is inherited by $g \circ f$ and by f^{-1}. □

This is the equivalence relation that canonicalization must decide. Atom maps, coordinates, toolkit CIP labels and ranks, provenance, and stability evidence are

excluded unless explicitly declared constitutional attributes. Missing or unsupported configured information fails closed instead of producing a partial identity certificate. Note that (S3) is a condition on *orbits*, not on stored tuples: by Lemma S9(i) it is invariant under the choice of representative at either end, so no local normalization is needed before the global comparison.

2.6.4 Faithful auxiliary-graph expansion

Let $\Phi_\chi(\mathcal{X})$ be a finite colored graph with the vertex roles in Table 4. Bonds are represented through incidence vertices, which retains bond/resource type and permits a uniform attachment model. Every virtual resource is owner scoped. Most importantly, a local configuration is encoded by its *complete orbit*, not by coloring one arbitrarily chosen representative.

Table 4 Colored auxiliary-graph roles.

Vertex role	Encoded information
Atom	constitutional atom attributes
Bond/resource	typed bond incidence or explicit Lewis resource
Stereo locus	support family and geometry
Port	owner-specific material or virtual ligand incidence
Tuple relation	one ordered tuple in the allowed local orbit
Position slot	tuple position, joining a tuple relation to its port
Path/plane relation	reversal, rotation, cyclicity, and orientation constraints

For a finite-frame locus s with representative u_s, construct

$$O_s = \{\gamma \cdot u_s : \gamma \in \Gamma_{\tau_s}\}.$$

For every tuple in O_s, Φ_χ contains one tuple-relation vertex. It is joined to the appropriate port through position-colored slot vertices. Slot colors preserve tuple position, while the set of all tuple vertices preserves exactly the permitted orbit. Hence a tetrahedral expansion admits all twelve even port actions and no odd one; a planar-bond expansion admits exactly its four V_4 frame symmetries.

Lemma 2 (Local orbit becomes a graph constraint)**.** *Let s be a finite-frame locus with frame u_s, whose k_s ports are distinct, and let π be a permutation of those ports. Write $\hat{\pi} = u_s^{-1} \circ \pi \circ u_s \in S_{k_s}$ for π expressed in slot coordinates. Then π extends to a color-preserving automorphism of the locus–tuple–slot gadget if and only if $\hat{\pi} \in \Gamma_{\tau_s}$.*

Proof. A tuple vertex is adjacent, through position-colored slot vertices, to exactly one port per position, so a color-preserving extension of π must send the vertex of a tuple $t \in O_s$ to the vertex of $\pi(t)$, where π acts entrywise. Such an extension exists if and only if $\pi(O_s) = O_s$, and it is then unique.

Entrywise relabeling commutes with the slot action by Lemma S1, so $\pi(O_s) = \{\pi(\gamma \cdot u_s) : \gamma \in \Gamma_{\tau_s}\} = \Gamma_{\tau_s} \cdot \pi(u_s)$. Moreover $(\hat{\pi} \cdot u_s)_i = u_s(\hat{\pi}(i)) = \pi(u_s(i))$, that is, $\pi(u_s) = \hat{\pi} \cdot u_s$. Hence

$$\pi(O_s) = O_s \iff \hat{\pi} \cdot u_s \in \Gamma_{\tau_s} \cdot u_s \iff \hat{\pi} \in \Gamma_{\tau_s},$$

where the last step uses the uniqueness of alignments in Lemma S2, which requires the entries of u_s to be distinct. If $\hat{\pi} \notin \Gamma_{\tau_s}$, then some allowed tuple is sent outside O_s, and since Φ_χ contains a tuple vertex for every element of O_s and for no other tuple, no color-preserving extension exists. □

Identifying ports with slots through u_s is the convention used in the rest of this section, so we write $\rho_s \in \Gamma_{\tau_s}$ for the condition $\hat{\rho}_s \in \Gamma_{\tau_s}$. The identification is harmless precisely because Lemma 2 is stated for the conjugate, and conjugation by a different representative of the same orbit changes $\hat{\pi}$ only within $\Gamma_{\tau_s} \hat{\pi} \Gamma_{\tau_s}$.

For tetrahedral stereo, Lemma 2 and Proposition S3 say that exactly the even port permutations extend. An odd alignment is not stored as a rejected bit; it maps the complete A_4 tuple set to the opposite coset, so the two local gadgets remain distinguishable inside the global graph.

Lemma 3 (Extended support relations)**.** *Within the declared admissible support domain, the path, plane and framework expansions preserve and reflect their configured support transformations.*

Proof. A cumulene or extended-cis/trans locus is incident with precisely its selected path-bond vertices. These edges form a simple path, so its ordered material sequence is recoverable up to reversal. The terminal frame orbit is anchored to its two endpoints through owner-specific ligand incidences. Applying Lemma 2 to that orbit therefore permits exactly the registered terminal action simultaneously with the selected path transport. No arbitrary permutation of internal path positions is introduced.

An open helical support encodes its path and reversal as complete ordered tuples; a cyclic support encodes all rotations of both traversals. Distinct path atoms make these tuples distinct. Position colors force an auxiliary isomorphism to transport the entire tuple set, which is exactly the declared reversal/dihedral support relation. The locus color retains helical parity, cyclicity and the declared coupling identity. A planar-chirality locus instead encodes the pilot followed by all rotations of one oriented plane traversal. The pilot slot is distinguished. Transport of this tuple set is exactly transport of the pilot and the oriented cycle; reversal changes the encoded orientation and cannot be silently admitted as a rotation.

A framework encodes its whole support through incidence vertices and each frame through a center-attached frame vertex. For every frame it encodes the twelve proper tetrahedral tuples at the configured parity $c_j\omega$, where c_j is its relative frame relation and ω the common supplied orientation. Frame centers are distinct. Lemma 2 forces one constitutional map to carry every frame orbit to its image simultaneously. This is exactly preservation of the configured coupled-frame relation; there is no independent choice to invert an individual frame. Distinct tuple entries, position colors, typed virtual owners and unique carrier keys determine the auxiliary extension of a given atom map in each construction. □

2.6.5 Faithfulness

Theorem 4 (Faithful stereograph expansion)**.** *For configured stereographs over the supported families, with distinct ports at every finite-frame locus and the admissible support and uniqueness conditions of the representation definition,*

$$\mathcal{X} \cong_{\mathrm{st}} \mathcal{Y} \quad \Longleftrightarrow \quad \Phi_\chi(\mathcal{X}) \cong_{\mathrm{col}} \Phi_\chi(\mathcal{Y}).$$

Moreover the correspondence $f \mapsto \Phi_\chi(f)$ below is a bijection between stereo-isomorphisms and color-preserving isomorphisms of the expansions.

Proof. ($\Rightarrow$) Let f be a stereo-isomorphism. Define $\Phi_\chi(f)$ role by role: on atom vertices it is f; on bond and resource incidence vertices it is the induced map on incidences, which is well defined by (S1); on locus and port vertices it is $s \mapsto f(s)$ and the port transport of (S2); on a tuple vertex $t \in O_s$ it is the vertex of $f(t)$, which lies in $O_{f(s)}$ because $\rho_s(f) \in \Gamma_{\tau_s}$ by (S3) and $f(\gamma \cdot u_s) = \gamma \cdot f(u_s)$ by Lemma S1; on the slot vertex joining t to its port in position i it is the corresponding slot vertex of $f(t)$, preserving the position color. Path and plane relation vertices are transported by (S4). All colors are preserved by construction, and adjacency is preserved because every auxiliary edge is defined from data that f transports.

($\Leftarrow$) Let F be a color-preserving isomorphism of the expansions. Vertex colors encode the role, so F maps atoms to atoms, loci to loci of the same family and geometry, ports to ports with the same owner scope, tuple vertices to tuple vertices, and position-i slots to position-i slots. Its restriction f to atom vertices is a bijection preserving all χ-colors and, through the incidence vertices, all typed bonds and Lewis resources; hence $f \in \mathrm{Iso}_\chi(G, H)$ and (S1) holds. Locus and port roles give (S2). At a finite-frame locus s, F maps the tuple set O_s bijectively onto $O_{f(s)}$, and by the slot colors this map is entrywise application of the port transport; therefore $\pi(O_s) = O_{f(s)}$ for the induced port bijection π, which by the computation in the proof of Lemma 2 is

equivalent to $\rho_s(f) \in \Gamma_{\tau_s}$. This is (S3). The declared reversal, rotation, and orientation tuple sets, and the coupled-frame relations, give (S4) by Lemma 3. Hence f is a stereo-isomorphism.

The two constructions are mutually inverse: $\Phi_\chi(f)$ restricts to f, and conversely an auxiliary isomorphism is determined by its restriction to atoms, because every non-atom vertex is uniquely identified by the atoms, positions, and colors it is attached to under the declared uniqueness conditions. In particular, duplicate carrier gadgets and parallel indistinguishable resource vertices are excluded. □

The theorem is the bridge missing from a collection of local descriptors. It makes the complete colored graph, rather than an iterative exchange of local ranks, the single object whose symmetries must be resolved.

2.6.6 Exact canonical certificate

Let Can be a canonical form for finite colored graphs, that is, a computable map satisfying

$$\mathrm{Can}(K) \cong_{\mathrm{col}} K, \qquad K \cong_{\mathrm{col}} K' \Longrightarrow \mathrm{Can}(K) = \mathrm{Can}(K'). \tag{7}$$

The two conditions together imply the biconditional $\mathrm{Can}(K) = \mathrm{Can}(K') \iff K \cong_{\mathrm{col}} K'$: the forward direction follows from the first condition and transitivity of $\cong_{\mathrm{col}}$, the converse is the second condition. This is the only property of the canonizer used below; no assumption is made about how the search is organized. The implemented native engine realizes Eq. (7) by exact equitable refinement, individualization of every unresolved cell, exhaustive exploration of the remaining branches, and selection of the least full color/adjacency serialization. The mathematical canonical-form function is total on finite colored graphs. A resource-bounded implementation is partial: a completed exact search returns that function's value; an interrupted search returns a refusal. The biconditional below applies to successfully returned exact values, not to refusal objects. To distinguish a representative graph from its encoded tuple, let $\mathcal{L}(K, \pi)$ denote the full

ordered color/adjacency serialization under a vertex bijection $\pi : V(K) \to [|V(K)|]$. We write $\mathrm{can}(K)$ for the lossless serialization of $\mathrm{Can}(K)$ in its canonical order. Thus $\mathrm{can}(K) = \mathrm{can}(K')$ exactly when $K \cong_{\mathrm{col}} K'$; the upper-case operator returns a graph, and the lower-case operator its encoding. An incomplete invariant need satisfy only the forward implication. Figure 6 illustrates why renumbering invariance alone cannot establish exact identity.

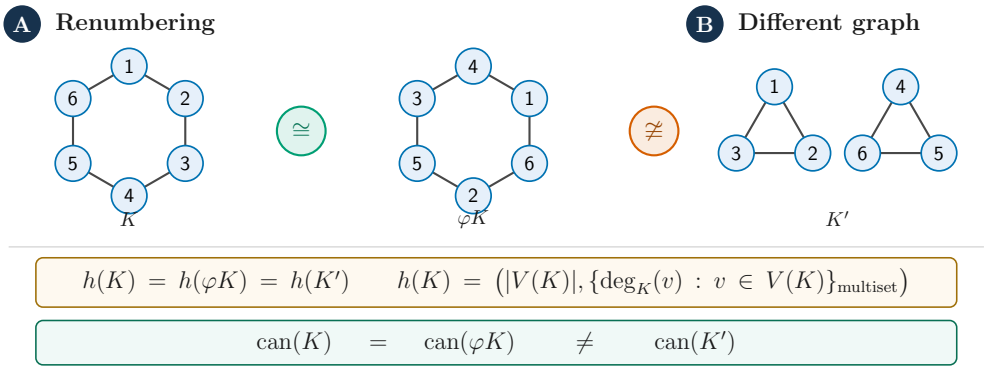


Fig. 6 Renumbering invariance and exact identity have different obligations. The first two colored graphs are isomorphic under the displayed renumbering. The third has two connected components and is not isomorphic to them. All three have six identically colored vertices, degree two at each vertex, and identical edge colors, so the displayed vertex-count/degree invariant collides. A full canonical encoding identifies the renumbered pair and separates the nonisomorphic graph. Numbers are display identifiers, not vertex colors. The example concerns colored-graph identity rather than stereochemical perception. The independent assignment-partition checks require both invariance within reference classes and separation across them.

Definition 2 (Canonical stereograph certificate)**.**

$$C^{\chi}_{\mathrm{SG}}(\mathcal{X}) = \mathrm{Can}(\Phi_{\chi}(\mathcal{X}))\,.$$

The identity profile χ, vertex colors, and graph serialization are part of the mathematical domain of Can. A concrete interchange encoding must version those choices unambiguously. Any accompanying digest is an index only; equality is decided on the full canonical object.

We suppress the superscript when the profile is fixed by context.

Theorem 5 (Canonical stereograph theorem)**.**

$$C_{\mathrm{SG}}(\mathcal{X}) = C_{\mathrm{SG}}(\mathcal{Y}) \quad \Longleftrightarrow \quad \mathcal{X} \cong_{\mathrm{st}} \mathcal{Y}.$$

Proof. By Theorem 4, $\mathcal{X} \cong_{\mathrm{st}} \mathcal{Y}$ is equivalent to $\Phi_\chi(\mathcal{X}) \cong_{\mathrm{col}} \Phi_\chi(\mathcal{Y})$, and by Eq. (7) the latter is equivalent to equality of the two canonical forms, which is the definition of $C_{\mathrm{SG}}(\mathcal{X}) = C_{\mathrm{SG}}(\mathcal{Y})$. □

Corollary 6 (Numbering independence)**.** *For every relabeling f, $C_{\mathrm{SG}}(f\mathcal{X}) = C_{\mathrm{SG}}(\mathcal{X})$. The certificate therefore satisfies both requirements of Eqs. (5) and (6).*

Proof. A relabeling is a stereo-isomorphism $\mathcal{X} \to f\mathcal{X}$: it satisfies (S1) and (S2) by construction, and $\rho_s(f) = \mathrm{id}$ for every locus because f transports each frame to the frame it defines on the image ports. Apply Theorem 5. □

The raw-to-formal factorization in Corollary S13 permits validation of the same global partition without repeating every local frame representative. Raw enumeration remains a separate executable check of construction and transport.

The certificate projects a canonical order for atoms, bonds/resources, stereo loci, and ligand ports. On symmetric inputs, original-handle orders are witnesses modulo a verified automorphism and are not falsely presented as a unique equivariant choice. The symmetry of the auxiliary graph is exactly the stereo-aware symmetry of the original object.

Under the admissibility assumptions of Theorem 4, restriction to atoms is an isomorphism from auxiliary automorphisms to stereo-preserving constitutional automorphisms. Proposition S14 proves that the projection loses no symmetry and adds none. Reported atom orbits therefore describe the configured object's actual symmetry. Complete group materialization is a distinct output cost; exact certificates do not require enumerating every group element.

2.6.7 Algorithm and complexity

Algorithm 1 is the constructive form of Theorem 5. The exact-canonizer call returns a full canonical graph K, a canonical map κ, automorphisms, and search evidence. Any node, depth, or time stop produces a refusal rather than a partial code.

Algorithm 1 Exact canonicalization of a configured molecular stereograph

Require: Configured stereograph $\mathcal{X} = (G, \mathcal{R})$
Ensure: Exact certificate, canonical orders, automorphisms, and search evidence

$G_0 \leftarrow \textsc{NormalizeConstitution}(G)$
$\textsc{RequireCompleteSupportedRegistry}(G_0, \mathcal{R})$
$A \leftarrow \textsc{InitializeColoredGraph}$
for all $v \in V(G_0)$ **do**
    $c_v \leftarrow \textsc{ConstitutionColor}(v)$
    $\textsc{AddAtom}(A, v, c_v)$
end for
for all $e \in E(G_0)$ **do**
    $\textsc{AddTypedBondIncidence}(A, e)$
end for
for all $s \in \textsc{CanonicalLocusInputOrder}(\mathcal{R})$ **do**
    $(\ell_s, P_s) \leftarrow \textsc{AddLocusAndOwnedPorts}(A, G_0, s)$
    if $\textsc{HasFiniteFrame}(s)$ **then**
        $O_s \leftarrow \{\gamma \cdot u_s : \gamma \in \Gamma_{\tau_s}\}$
        for all $t \in \textsc{CanonicalTupleOrder}(O_s)$ **do**
            $\textsc{AddTupleAndPositionSlots}(A, \ell_s, P_s, t)$
        end for
    else
        $\textsc{AddPathOrPlaneRelations}(A, \ell_s, P_s, s)$
    end if
end for
$(K, \kappa, \mathcal{A}, \eta) \leftarrow \textsc{ExactColoredCanon}(A)$
if $\eta.\textsc{stopped}$ **then**
    return $\textsc{Refusal}(\eta.\text{reason})$
end if
$C \leftarrow \textsc{SerializeExactly}(K, \chi)$
$\Omega \leftarrow \textsc{ProjectOrdersAndAutomorphisms}(\kappa, \mathcal{A})$
return $(C, \Omega, \kappa, \eta)$

For a fixed finite-frame locus s, the expansion contributes one locus vertex, k_s ports, $|\Gamma_{\tau_s}|$ tuple vertices, and $k_s|\Gamma_{\tau_s}|$ position slots, plus any private virtual-resource vertices. This fixed-arity term is $1 + k_s + (1 + k_s)|\Gamma_{\tau_s}|$. The catalogue also contains

supports of variable size, whose costs must be counted separately. An open helical path of length L contributes $1+2+2L$ vertices; a cyclic path contributes $1+2L+2L^2$. A planar-chirality cycle of length L with a pilot contributes $1+L+L(L+1)$. These follow directly from the explicit tuple construction: two open traversals, $2L$ cyclic traversals, and L oriented plane traversals of width $L+1$, respectively.

A coupled framework with h support atoms, r tetrahedral frames and ν virtual references counted across frames contributes $1+h+61r+12\nu$ vertices: one locus, support incidences, and for each frame one frame node, twelve representations and forty-eight slots. Path-supported cumulene and extended-cis/trans carriers additionally contribute a term linear in their path length. Consequently the expansion has size

$$O\left(n+\sum_{s\in S_{\text{finite}}} k_s|\Gamma_{\tau_s}|+\sum_{s\in S_{\text{path}}} L_s+\sum_{s\in S_{\text{cycle}}\cup S_{\text{plane}}} L_s^2+\sum_{s\in S_{\text{framework}}} (h_s+r_s+\nu_s)\right),$$

where n includes constitutional vertices and edges. Construction is linear in the expansion's output size. Only the fixed-arity catalogue with bounded support multiplicity yields a bound linear in molecular size; fixed finite geometry types do not bound cycle or plane length. Certificate comparison is linear in certificate length. Complete graph canonical search has no polynomial-time guarantee, and resonance enumeration, assignment products, and materializing a complete automorphism group can also be superlinear or have exponentially large output. They are separate costs, with explicit refusal conditions in a bounded implementation. The present implementation retains the exhaustive search as a correctness oracle and uses verified automorphism pruning in the production exact path; Section 3.2 reports the remaining scalability boundary.

2.6.8 Consequences for identity and matching

Two equal certificates prove identity. Unequal hashes do reject identity, but equal hashes do not replace comparison of the full certificate. This contract supports exact stereoisomer deduplication and stereo-aware atom/ligand orbits. It also supplies a terminal check for local neighbor-alignment and stereo-morphism witnesses: a structural mapping is accepted only when every configured locus transports into the same local orbit under the declared information and presence policies. Identity witnesses compose, while query wildcards remain outside stored equality. The complete morphism policy matrix and composition proofs are in Appendix J.

2.7 Molecular chirality

2.7.1 Mirror action and exact classification

Mirror action is geometry specific, and for a finite-frame family it is fixed by which improper spatial operation the geometry actually realizes on its slots.

Definition 3 (Improper extension and local mirror)**.** For a finite geometry τ, let $\widehat{\Gamma}_\tau$ with $\Gamma_\tau \trianglelefteq \widehat{\Gamma}_\tau \leq \Sigma_\tau$ be the *improper extension*: the permutation group generated by Γ_τ together with the slot action of the reflections of the idealized polyhedron. The registered index is $\left[\widehat{\Gamma}_\tau : \Gamma_\tau\right] \in \{1, 2\}$, and the local mirror is

$$m_\tau([u]_{\Gamma_\tau}) = \begin{cases} [\nu \cdot u]_{\Gamma_\tau}, & \left[\widehat{\Gamma}_\tau : \Gamma_\tau\right] = 2, \quad \nu \in \widehat{\Gamma}_\tau \setminus \Gamma_\tau, \\ [u]_{\Gamma_\tau}, & \left[\widehat{\Gamma}_\tau : \Gamma_\tau\right] = 1. \end{cases}$$

Path- and plane-supported families instead use their declared sense inversion on the canonical form.

Lemma 7 (The local mirror is a well-defined involution)**.** *m_τ does not depend on the choice of ν, commutes with reference relabeling, and satisfies $m_\tau \circ m_\tau = \mathrm{id}$. If moreover $\widehat{\Gamma}_\tau = \Sigma_\tau$ and $[\Sigma_\tau : \Gamma_\tau] = 2$, then m_τ coincides with* opp *of Corollary S7.*

Proof. The index-two case is Corollary S7 applied to the pair $\Gamma_\tau \trianglelefteq \widehat{\Gamma}_\tau$ instead of $\Gamma_\tau \leq \Sigma_\tau$: the nonidentity coset is unique, so $[\nu \cdot u]$ is independent of ν, and applying it twice returns Γ_τ. The index-one case is the identity map. Commutation with relabeling is Lemma S1. For the final claim, the added hypotheses give $\widehat{\Gamma}_\tau \setminus \Gamma_\tau = \Sigma_\tau \setminus \Gamma_\tau$, so the two constructions select the same coset. □

Instantiating Definition 3 explains the chemistry case by case. Tetrahedral geometry has $\widehat{\Gamma} = S_4 \supset A_4$ of index two, so the mirror is the enantiomeric orbit; the atrop and cumulene axes likewise have index two and exchange P with M. Trigonal-bipyramidal and octahedral geometries have index-two improper extensions inside Σ_τ, namely the images of D_{3h} and O_h, so the mirror is the registered inversion of their 20 and 30 configurations rather than a binary flip. Square-planar and planar-bond geometries realize their reflection in the molecular plane, which fixes every slot, so the index is one and the mirror is the identity. In particular E and Z are distinct configurations but are *not* mirror images, even though they form a binary pair: being an index-two pair under Σ_τ and being exchanged by an improper operation are different statements, and only the second one is chirality.

Writing μ_B for the map that applies m_{τ_s} at every configured locus of $\mathcal{X}$ inside the boundary B while leaving the constitution unchanged, Lemma 7 gives the two laws used below,

$$\mu_B^2 \mathcal{X} = \mathcal{X}, \qquad f_*(\mu_B \mathcal{X}) = \mu_B(f_* \mathcal{X}), \tag{8}$$

where the subscript records the support boundary over which the configured object is complete. For path and plane families these laws are requirements on the declared sense inversion rather than consequences of Lemma 7.

Definition 4 (Molecular chirality)**.** Let $\mu\mathcal{X}$ be the configured stereograph obtained by applying the geometry-specific mirror action while leaving constitution unchanged. $\mathcal{X}$ is achiral exactly when $\mathcal{X} \cong_{\mathrm{st}} \mu\mathcal{X}$, and chiral otherwise. By Eq. (8) this does not depend on the numbering of $\mathcal{X}$ or on the stored local representatives.

Corollary 8 (Exact mirror criterion)**.**

$$\mathcal{X} \textit{ is achiral} \quad \Longleftrightarrow \quad C_{\mathrm{SG}}(\mathcal{X}) = C_{\mathrm{SG}}(\mu\mathcal{X}).$$

Proof. Definition 4 states achirality as $\mathcal{X} \cong_{\mathrm{st}} \mu\mathcal{X}$, and Theorem 5 converts that relation into equality of the two certificates. Both objects are complete over the same boundary, so the certificate of $\mu\mathcal{X}$ is defined whenever that of $\mathcal{X}$ is. □

Canonicalization already sends every relabeling $f_*\mathcal{X}$ to the same class as $\mathcal{X}$, whether the object is chiral or achiral. Achirality asks whether the mirror joins that class as well. For complete supported input,

$$|\{C_{\mathrm{SG}}(\mathcal{X}), C_{\mathrm{SG}}(\mu\mathcal{X})\}| = \begin{cases} 1, & \mathcal{X} \text{ achiral}, \\ 2, & \mathcal{X} \text{ chiral}. \end{cases}$$

Figure 7 makes the scope of this comparison explicit across six chemical cases and five carrier geometries. The mirror operation is applied to every configured carrier at once; canonicalization then searches for one global stereo-isomorphism. It is therefore neither an enumeration of atom numberings nor a test that independently cancels local configurations.

Algorithm 2 turns this criterion into a proof-bearing decision. When the certificates agree, the two canonical maps construct the mirror isomorphism rather than merely asserting that one exists.

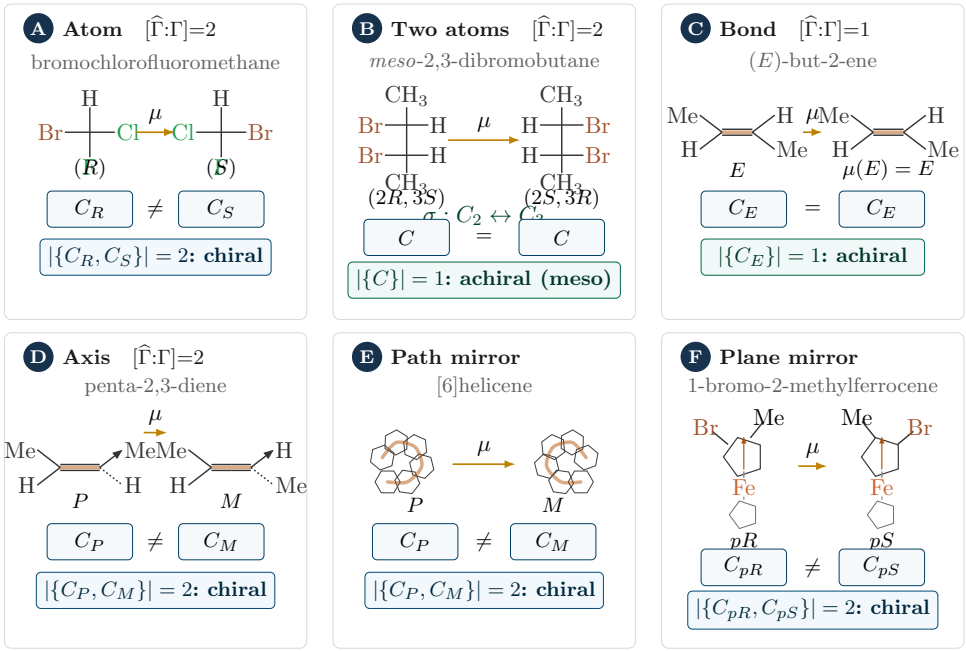


Fig. 7 Whole-molecule mirror canonicalization in six real chemical cases. Each panel header gives the support and the improper index $\left[\widehat{\Gamma} : \Gamma\right]$ of Definition 3, which decides whether μ can move the local state at all; the verdict counts distinct certificates by Corollary 8. Fischer projections place horizontal bonds toward the viewer and vertical bonds away. (A) R/S-bromochlorofluoromethane gives two certificates. (B) For *meso*-2,3-dibromobutane the index is two at both centers, yet the center-exchange automorphism σ identifies the mirror and one certificate remains: compensation is a property of the global quotient, not of the local index. (C) The planar bond has index one, so spatial reflection preserves the E state of but-2-ene and does not manufacture Z. (D) P/M-penta-2,3-diene, (E) P/M-[6]helicene, and (F) pR/pS-1-bromo-2-methylferrocene each retain two mirror-related certificates, the last two through the declared sense inversion of a path and of an oriented plane. One global comparison therefore covers atom, bond, axis, path, and plane carriers as well as meso compensation. Arbitrary atom renumberings within each stereograph have already been quotiented by canonicalization.

This criterion handles internal compensation directly. A *meso* structure is achiral because an exact atom mapping carries its complete configured graph onto the mirror, not because a heuristic cancels a list of local labels. In the language of Section 2.6.2, an achiral assignment is a fixed point of the mirror composed with the action of A. The same comparison classifies pairs.

Definition 5 (Stereoisomer relations)**.** Let $\mathcal{X}$ and $\mathcal{Y}$ be complete configured stereographs whose constitutions are χ-isomorphic and whose unconfigured carrier inventories correspond under a constitutional isomorphism. They are *identical*

Algorithm 2 Exact whole-molecule chirality by mirror canonicalization

Require: Configured stereograph $\mathcal{X} = (G, \mathcal{R})$
Ensure: ACHIRAL, CHIRAL, INCOMPLETE, or UNSUPPORTED

$q \leftarrow \textsc{RegistryCompleteness}(\mathcal{X})$
if $q \neq$ COMPLETE **then**
    return q with its unsupported or missing-locus evidence
end if
$r \leftarrow \textsc{CanonicalStereoGraph}(\mathcal{X})$
if $\textsc{IsRefusal}(r)$ **then**
    return r
end if
$\overline{\mathcal{X}} \leftarrow \textsc{GeometrySpecificMirror}(\mathcal{X})$
$\bar{r} \leftarrow \textsc{CanonicalStereoGraph}(\overline{\mathcal{X}})$
if $\textsc{IsRefusal}(\bar{r})$ **then**
    return $\bar{r}$
end if
if $r.C = \bar{r}.C$ **then**
    $\psi \leftarrow \bar{r}.\kappa^{-1} \circ r.\kappa$
    $w \leftarrow \textsc{ProjectAtomIsomorphism}(\psi)$
    return $\textsc{Achiral}(r.C, w)$
else
    return $\textsc{Chiral}(r.C, \bar{r}.C)$
end if

when $C_{\mathrm{SG}}(\mathcal{X}) = C_{\mathrm{SG}}(\mathcal{Y})$; *enantiomers* when $C_{\mathrm{SG}}(\mathcal{X}) = C_{\mathrm{SG}}(\mu\mathcal{Y})$ but $C_{\mathrm{SG}}(\mathcal{X}) \neq C_{\mathrm{SG}}(\mathcal{Y})$; and *diastereomers* otherwise. Incompatible represented inventories return an information mismatch, rather than a chemical stereoisomer relation.

Lemma 9 (The three relations are symmetric and exhaustive)**.** *Exactly one of the three cases holds for each such pair, and each is a symmetric relation. Moreover $\mathcal{X}$ and $\mathcal{Y}$ are enantiomers if and only if $\mu\mathcal{X}$ and $\mathcal{Y}$ are identical and $\mathcal{X}$ is chiral.*

Proof. Exclusivity and exhaustiveness are immediate from the definition, which distinguishes cases by two certificate equalities. For symmetry it suffices to treat the enantiomer case. By Theorem 5, $C_{\mathrm{SG}}(\mathcal{X}) = C_{\mathrm{SG}}(\mu\mathcal{Y})$ means $\mathcal{X} \cong_{\mathrm{st}} \mu\mathcal{Y}$. Applying μ and using naturality in Eq. (8) gives $\mu\mathcal{X} \cong_{\mathrm{st}} \mu^2\mathcal{Y} = \mathcal{Y}$, which is the same statement with the roles exchanged; the inequality clause is symmetric for the same reason. The final claim restates the two conditions using $\mu^2 = \mathrm{id}$ and Corollary 8. □

Because C_{SG} is numbering independent (Corollary 6) and μ is natural, all three verdicts are invariant under relabeling of either input and independent of which local representatives were stored.

2.7.2 Complete inputs and completion populations

The end-to-end operation normalizes the declared constitution, obtains typed carrier evidence, validates and attaches supplied configuration, audits the requested support boundary, and performs exact mirror comparison. Every required unresolved locus or unsupported information state refuses a binary answer. The conditional pipeline theorem in Appendix A.7 requires equivariant preparation, sound attachment, a correct boundary audit, faithful expansion and completed exact searches. It concerns the represented configured object; it does not establish complete physical annotation or experimental persistence.

A completion assessor instead quantifies over supported assignments consistent with partial evidence. Its population outcomes are necessarily achiral, necessarily chiral, configuration dependent, and unsupported/incomplete. Two validated opposite-class witnesses establish configuration dependence without exhaustion. A one-sided answer requires exhaustive coverage of a nonempty population or an exact proof excluding the opposite class. A reached cap alone cannot supply that proof. Empty or unresolved populations refuse. Appendix S11 gives the full algorithm and conditions.

The American Chemical Society (ACS) source-specific assessor additionally uses topology probes for globally stereogenic cases not localized by the source registry. A separate exact-plus-global protocol represents those probes as an orientation-unspecified coupled framework: topology may establish necessary chirality without choosing one enantiomeric orientation. Configured framework orientation still requires explicit authorized evidence. These procedures have different information boundaries and are evaluated separately from the source-declared certificate.

Enhanced-stereo populations and configurational stability remain outside the configured-identity theorem. A coordinate record can supply configuration only under its declared transport and geometry rules. A constitutional axis can be structurally stereogenic while interconverting rapidly; the identity model does not infer an experimental lifetime from the existence of that axis.

2.8 Validation datasets and protocols

The evaluation is divided by scientific question (Table 5). No pooled "stereo accuracy" is reported. Designed algebraic fixtures test exact laws; public data are used only for the task and annotation level they actually provide.

Figure 8 summarizes the evidence under its separate identity, perception, naming and chirality contracts.

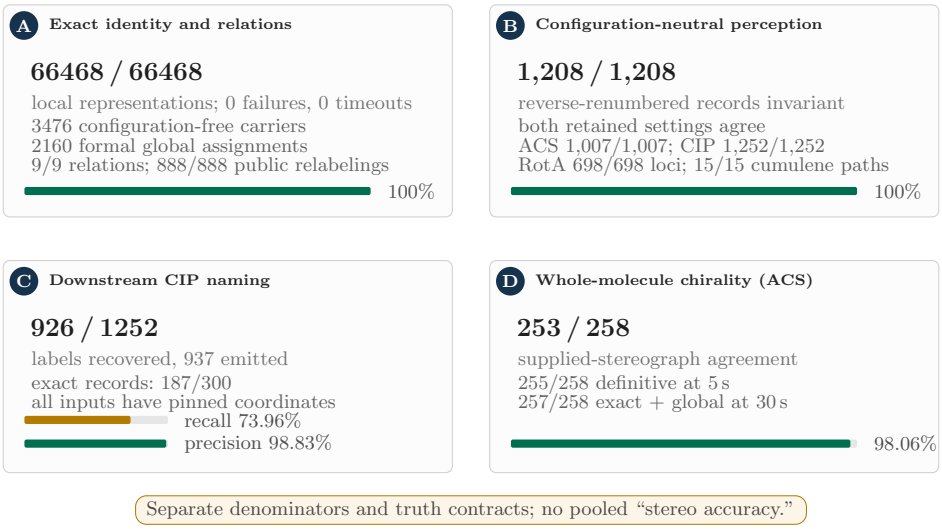


Fig. 8 Task-separated overview of the frozen benchmark evidence. Each tile carries one hero ratio, its own denominator, and its own truth contract: exact identity and relation laws (A), carrier perception (B), downstream CIP naming (C), and ACS whole-molecule chirality (D). Every meter is printed with its numeric rate, and each is read against its own tile only. Panel C reports 187/300 as exact-record accuracy, alongside label recall and precision. All 300 inputs have pinned coordinates; 187 is the number with an exactly correct predicted label set. The four tiles are reported side by side but are never pooled into one accuracy statistic.

Table 5 Experiments and their claims.

Track	Protocol	Claim tested
Local algebra	Exhaust every registered finite action; test path/plane involutions and witness composition	Local configuration and transport laws
All-family canonicalization	940 local representations, 67 expected classes, per-class and representative relabelings	Faithful local-orbit encoding and numbering-independent exact certificates
Configuration-free local	All 66,468 raw representations of 3,476 carriers over 1,218 Internal, ACS, CIP, and RotA rows	Exhaustive collapse to formal local classes under fixed molecular constitution
Mixed-family global	Formal A/B/C over 30 cases and 2,160 assignments; raw A/B and raw C/VS146 over 3,353,600 representations	Simultaneous local-state composition and exact raw-to-formal-to-global factorization
Public permutation audit	All $5!+4!+4!+6! = 888$ atom permutations of four configured ACS molecules	Certificate invariance on public structures
Multi-element composition	All 32 binary assignments on six two- and three-element graphs; seven additional ACS molecules	Global symmetry, mirror closure, and coupled local states
Pairwise relations	9 designed identity, enantiomer, diastereomer, constitution, and incomplete cases	Whole-stereograph classification control flow
Perception	All 1,208 ACS, CIP, and RotA records under retained-neighbor-frame and erased-configuration settings	Carrier recovery and renumbering invariance, not handedness or stability
Derived naming	Pinned 300-record CIP suite	Rules 1a/1b/2 label coverage
Molecular chirality	258 ACS records, exact certificate and specialized topology-completion paths reported separately	Whole-molecule classification and current exact-engine coverage
New coordination examples	40 initial source geometries; 38 certified, 2 refused	Independently annotated configured source/mirror identity
Sourced negative controls	48 new biaryl constitutions	Replayed typed symmetry witnesses and negative perception
RotA structural review	60 frozen source-assisted loci	Independent per-case partitions and category-specific perception

The three external datasets remain separate. ACS `StereoMolGraph` provides 258 whole-molecule chiral/achiral labels. ChiralFinder RotA provides 650 positive-only axial records. The CIP validation suite provides 300 structures with 1,252 recommended local positions and labels. Its original publisher supporting archive is provided under CC BY-NC 4.0 [6]. Two citation punctuation repairs reproduce the evaluated SMILES bytes without changing scientific fields. Original and later amended coordinates have separate checksums. The primary naming report uses the original publisher coordinates; all 300 native and RDKit comparator records agree with the historical amended-coordinate evaluation.

These are overlapping development corpora. Under stereo-neutral RDKit canonical representation with isotope and charge identity retained, the 1208 rows contain 904 distinct constitutions: 214 in ACS, 255 in CIP, and 650 in RotA. All 214 ACS constitutions occur in CIP; one CIP constitution also occurs in RotA. These are representation-level counts without resonance normalization, not independent holdout examples.

For perception, the *retained* setting permits supplied configuration only as oriented neighbor frames at other elements. The focal frame, CIP properties, canonical local descriptors, and coordinates are excluded. The *erased* setting removes all supplied configuration. Every record is also evaluated after reverse atom renumbering. RotA additional candidates cannot be scored as false positives because the corpus is positive-only.

For the exact ACS supplied-configuration audit, each record has a 5 s fail-closed budget. Every explicit source `@`/`@@` atom configuration is retained even when the input toolkit's default cleanup would discard it; additional undeclared potential loci remain unconstrained. Thus this row tests the source-declared configured certificate and makes no completeness claim. The stereo-stripped completion experiment uses a 256-isomer cap and measures the population of possible configurations; it does not claim to recover

erased stereo. When a tetrahedral assignment product exceeds the cap, exact XOR constraints over the constitutional automorphisms test whether any achiral completion exists, avoiding a raw 2^n search.

3 Results

3.1 Independent assignment-partition validation

Invariance alone cannot detect a certificate that merges every configuration. We therefore compare complete assignment partitions with a direct reference construction that does not use production canonical forms, stereo-isomorphism, the auxiliary encoder or production mirror operations for its truth. The reference derives proper A4/D4, TBP and octahedral actions from separately specified permutations or integer-coordinate geometry; terminal, path, plane and coupled-frame actions are specified independently from their raw fields. It exhausts attributed constitutional automorphisms that preserve a frozen typed carrier inventory, transports every formal assignment, and constructs its global assignment orbits directly.

The declared catalogue contains 33 constructed fixtures and 30 public A/B/C fixtures. All 63 cases and 2439 formal assignments agree with the complete production partition, compared as sets of assignment sets rather than by class count. Every production assignment is also mirrored and offset-renumbered. Independent mirror classes, mirror commutation and declared composition checks agree; all 2,439 retained atom-map witnesses replay and preserve typed constitutional colors. The public fixtures validate assignments conditional on the frozen perceived inventory. The constructed fixtures are mathematical tests, not an independent chemical holdout or a test of carrier perception.

The harness rejects 7 controlled defects: a constant certificate, erased local parity, an omitted locus, literal atom indices in identity, an incorrect planar-bond mirror,

prohibited equivalent-carrier exchange, and reversed noncommuting transport composition. Existing local and global benchmark gates now require agreement with these independently defined partitions, so an invariant but arbitrary merging cannot pass as a symmetry quotient. Full certificates are compared byte for byte in a temporary disk store; finite digests select candidates and do not define equality.

Reference limits are 50,000 constitutional automorphisms, 1,024 formal assignments, ten seconds per case and a one-GiB process memory guard. Production certificates have a five-second limit per computation. No case in this declared run reached an oracle or production limit. A refusal is never a pass. The largest case, VS126, has 1,024 assignments and took 1,100.87 seconds for the production, mirror and renumbering comparisons together. Maximum worker peak resident set size (RSS) was 134.08 MiB; its full-certificate disk store reached 1,203.12 MiB. These are observations from correctness validation, not a controlled speed comparison. The archived run began from clean commit `e2526b9`; source hashes remained unchanged throughout.

3.2 Canonicalization

The revised local and formal matrices compare complete production-certificate partitions with independent reference truth. Across all four local sources, 66468 raw representations of 3476 carriers complete; 6999 formal local states yield 5885 whole-graph classes, with 1114 local-to-global symmetry quotients. Every carrier passes the complete partition gate. Reference truth exhausts proper local frames and exact rooted constitutional transport queries rather than materializing the entire constitutional symmetry group. Every positive merge has a replayable typed atom map; all 6999 maps were audited. Ten-second, 1 GiB and separately declared 50,000-count query limits yield refusals; none occurred in this full run. The ten-family designed subset retains 67 classes. These results are conditional on the frozen carrier inventories, not independent validation of carrier perception.

The simultaneous mixed-family A/B/C matrix uses the separate exhaustive constitutional-automorphism reference: all 2160 formal assignments over 30 cases pass. These current formal results must be distinguished from the historical raw-representation runs below. The historical raw A/B and C/VS146 products exhaust 3,353,600 redundant encodings and agree with their production formal partitions. These are additional development checks, rather than independent negative-separation truth. Appendix B.5 preserves their full scope and the secondary relation gates. The submission's independent identity evidence is the complete-partition validation above.

3.3 Controlled costs and output size

A prospectively fixed study contains 90 inputs: all 33 independent fixtures, 24 unique ACS constitutions selected by input atom-count strata, 24 variable-support cases, three repeated-component stresses, three boundary probes and three resonance-profile examples. Three repetitions in fresh single-threaded workers compare generator output with complete-group output. Ordinary/stress workers have 30/120-second wall budgets, a 1 GiB native RSS limit and a 64 MiB report-output cap. All 540 planned trials were attempted. Source hashes remained unchanged at commit `e2526b9`; unrelated background contention is disclosed through trial load and CPU measurements.

Table 6 distinguishes successful computation from expected boundary refusal and resource censoring. The 513 completed trials have no certificate disagreement. Both output modes complete on 85 inputs; 86 have at least one complete code. Four identical tetrahedral components complete in generator mode, while full-group output reaches the memory limit in all repetitions. Eight components time out in generator mode and reach the memory limit in full-group mode. These outcomes remain part of coverage and are excluded from completed-only runtime summaries.

Figure 9 separates search from witness output and support expansion. At support length 33 the open-path, cyclic-path and plane constructions show the predicted

Table 6 All planned cost trials, including deliberate boundary probes. Resource censoring does not count as a completed identity decision.

Outcome	Generators	Full group
Complete	258	255
Unknown configuration	3	3
Unsupported inventory	3	3
Search-node refusal	3	3
Wall timeout	3	0
Memory limit	—	6
Planned	270	270

linear versus quadratic growth. Among completed generator trials, operation median is 0.029 s and search median is 0.006 s. These describe the selected heterogeneous workload, not deployment latency. Peak RSS includes interpreter and native libraries; its completed generator maximum is 365.9 MiB. The largest completed certificate is 34945023 bytes. Appendix D gives memory, output-size and phase measurements. Pairwise mirror checking dominates the completed four-component stress workload; full-group output is therefore only one possible bottleneck.

3.4 Registration demonstration

Independently labelled collections mix duplicates under proper frame moves and renumbering with distinct stereoisomers across six conventional frame families. At sizes 8, 16, 32, 64 and 128, each pipeline is executed three times with alternating order. Pairwise grouping compares records with existing group representatives; certificate grouping computes one full certificate per record and uses a digest index followed by exact byte equality. Both include input preparation and grouping/storage costs. All 30 pipelines recover the complete expected partitions. Separate chemical/Lewis/ACS profile checks and strict partial-input refusal pass.

Figure 10 shows no runtime crossover in this tested range. At 128 records, pairwise and certificate medians are respectively 0.140 s and 1.078 s. This small curated from-scratch workload supports registration correctness, not an acceleration claim or a

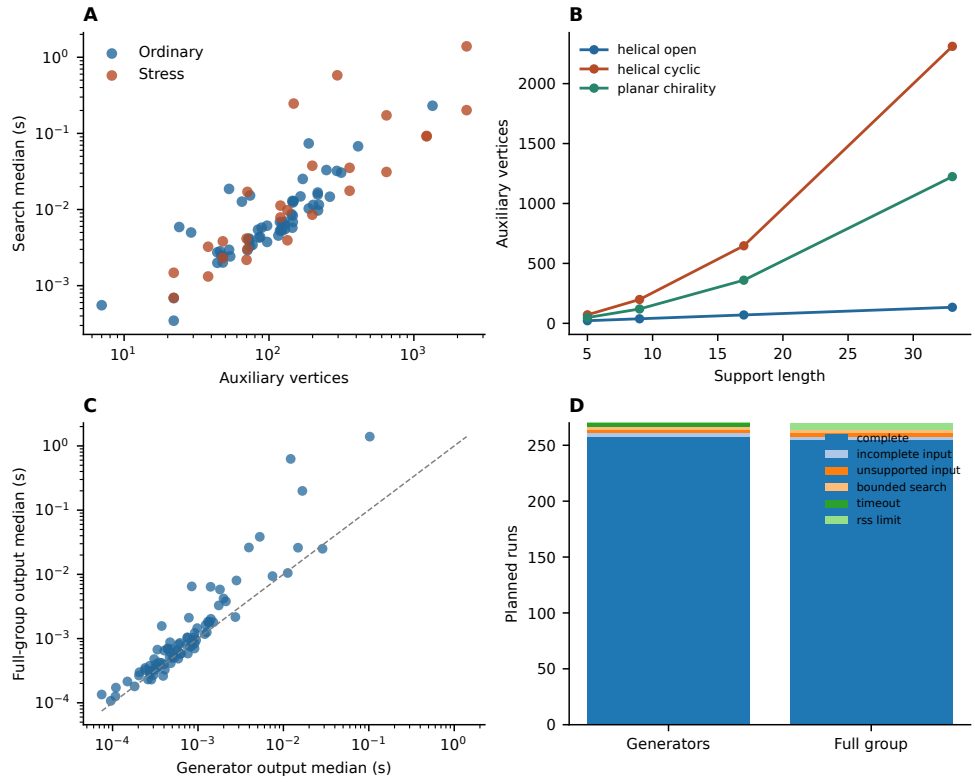


Fig. 9 Controlled configured-graph costs. (A) Generator-mode search medians against expansion size for completed cases. (B) Variable-support expansion size. (C) Witness-output medians on inputs completed by both modes; the dashed line indicates equality. (D) All planned trial outcomes. Each input has three repetitions; the plot and timings do not treat censored trials as completed. Different completion sets preclude interpreting pooled mode medians as a matched speed comparison.

chemical holdout claim. Memory is measured for the whole study rather than isolated pipelines, and background host load limits timing generalization. Persistent-index query workloads remain unmeasured.

3.5 New chemical examples and structural adjudication

We froze source selection and annotation before production evaluation. The new coordination inputs are initial generated geometries from the MACE archive [20], distributed under CC BY 4.0. They cover 20 Mn/Ru constitutions absent from the development corpora. Source donor isotopes encode coordination slots and are removed from physical isotope identity; supplied local tetrahedral frames are retained.

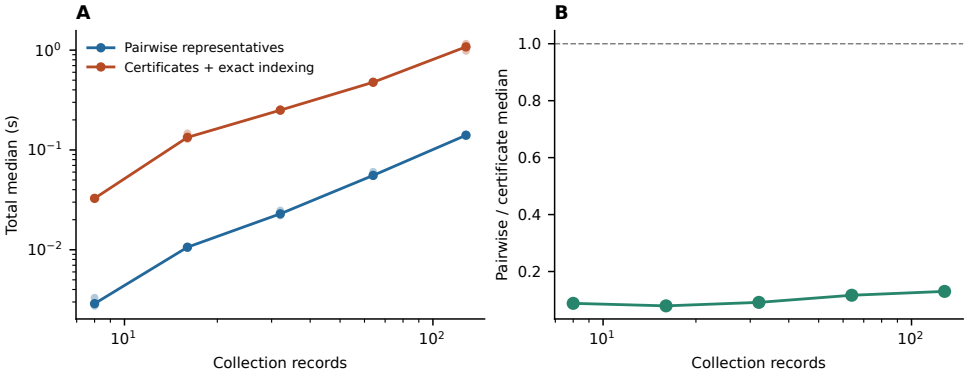


Fig. 10 Curated registration workload. (A) Total medians and individual repetitions for both pipelines. (B) Pairwise/certificate median ratio; values below one favour pairwise grouping. Both methods recover the independently specified groups, but certificate grouping is slower at every tested size.

An independent proper-rotation/reflection fit, followed by direct assignment-partition enumeration, supplies the reference. The frozen fit criteria certify 38/40 geometries and refuse 2 distorted examples; these refusals receive no production score.

All certified source and coordinate-reflected states agree with production mirrors and reverse renumbering in 152 full-certificate computations. The observed 47 source/mirror classes reproduce both directions of reference equality. Complete independent references contain 900 formal assignments and 230 classes across the selected constitutions; production coding covers the certified source/mirror states, rather than all reference assignments (Table 7). The source geometries do not establish experimental isolation or stability.

Separately, 48 new PubChem-sourced biaryl constitutions carry independently verified full typed automorphisms that fix each axis and exchange a terminal ligand pair. Original and reverse-renumbered perception agree with every negative witness. These are constitutional specificity controls, with no barrier or persistence annotation.

The RotA review fixes 60 historical additional loci across 26 strata and 48 source molecules. Algorithm confirmation/rejection status is withheld during annotation, but source supports and categories assist the review; it is development-corpus adjudication. Complete per-case partitions, mirror maps and renumberings agree over 128

assignments in 384 certificate computations. Of the 52 axis supports, 45 are ordinary single bonds and 7 are aromatic edges. The latter are conditional configuration-model tests, with no claim of ordinary single-bond atropisomerism. The remaining loci are 4 atom pairs and 4 spiro centers. Category-specific perception is checked separately from global configured identity; terminal ligand equivalence can coexist with distinct coupled configurations. All studies use automated independent mathematical annotation, rather than an independent human chemical review.

Table 7 Chemical evidence and its evaluated population. Reference classes are reported per constitution or case; counts are never pooled as chemical accuracy.

Protocol	Selected	Complete	Truth and evaluated scope
Coordination	40	38	Independent geometry and partitions; source/mirror states only; 2 geometry refusals
Sourced negatives	48	48	Full typed axis-fixed symmetry; 96 original/renumbered perception calls
RotA review	60	60	128 assignments; 98 classes summed per case; 120 perception calls

3.6 Perception and derived naming

Table 8 separates configuration-neutral carrier recovery from downstream naming. Retaining oriented frames at neighboring elements recovers every annotated locus. Erasing all configuration leaves 547 unresolved carbon tetrahedral positions whose center stabilizer contains an odd slot permutation. Every one is recovered in the retained setting, showing that the deficit is orientation-dependent ligand distinction rather than unsupported carrier topology.

The ACS and CIP erased misses number 264 and 283, respectively, and are all carbon tetrahedral carriers classified as symmetry related under the exact configuration-erased stabilizer. Axis recovery asserts typed candidate topology, not handedness or configurational stability. Additional candidate inventories are retained per record but are not presented as specificity scores where negative truth is absent. We therefore add

Table 8 Broad typed-carrier recovery in two settings.

Task	Retained	Erased	Reference
ACS configured source loci	1007	743	1007
CIP local positions	1252	969	1252
tetrahedral	980	697	980
planar	232	232	232
axial/helical	40	40	40
RotA typed loci	698	698	698
RotA cumulene paths	15	15	15
Reverse-renumbered records	1208	1208	1208

a separate 24-case synthetic negative protocol: 13 axis-like constitutions are rejected by explicit axis-fixed automorphisms that exchange a terminal ligand pair, and 11 contain no supported axis topology. All cases pass reverse atom renumbering and none is a confirmed-axis false positive. This protocol establishes constitutional specificity only; it supplies no experimental barrier, stability, or handedness truth.

With the pinned three-dimensional orientation input and supplied family categories selecting the projections, the CIP evaluator is exact on 187/300 records and recovers 926/1252 reference labels while emitting 937 (73.96% recall, 98.83% precision). It emits 11 wrong labels and misses 326. The SMILES-only baseline remains 175/300. Reviewed primary limitations among the 113 non-exact records are 83 ranking defects, 21 missing configured orientations, and 9 unsupported classes. The implementation includes Rules 1a/1b/2, simple mancude-ring averaging, Rule 3, and the reference-independent one-pair subset of Rules 4c/5; multi-unit Rule 4b remains fail-closed. These naming results do not participate in any canonical certificate. Moreover, complete assignment is not identifiable from the supplied SMILES contract: VS010 and VS011 have byte-identical input and mutually opposite helical label sets. Thus any deterministic SMILES-only method has an upper bound of 299 exact records. The pinned three-dimensional file resolves this input obstruction: the pair has opposite signs under the same translation- and proper-rotation-invariant oriented-volume witness. The coordinates distinguish this otherwise identical-input pair. They complete 2/2 helical and 5/5 CT4 records, while unambiguous transport completes 4/7 AT records. The remaining sequence rules,

three ambiguous AT transports, and TH3/TH5 projection remain algorithmic future work.

3.7 Molecular chirality

The ACS results expose three distinct protocols (Table 9). The specialized topology-completion classifier reproduces all 258 manual labels, matching the published `StereoMolGraph` result. It uses provisional probes and the publisher-compatible identity convention. The exact-stereograph row instead measures the configured certificate currently implemented and must not inherit the 258/258 score.

Table 9 Whole-molecule chirality on 258 ACS records. "Agreement" compares the declared procedure with the ACS topology labels; the exact-stereograph cross-profile differences are not internal certificate errors.

Procedure	Agreement	Fraction	Boundary
Topology completion (this work)	258/258	1.000	ACS-specialized probes
Exact + necessary global certificate	257/258	0.996	30 s budget; orientation unspecified
`StereoMolGraph` (published)	258/258	1.000	Published comparator
`StereoMolGraph` (pinned live replay)	254/258	0.984	Separate release/environment replay
InChI (published)	238/258	0.922	Published comparator
`RDKit` SMILES (published)	235/258	0.911	Published comparator
chython (published)	210/258	0.814	Published comparator
Exact supplied stereograph	253/258	0.981	255 definitive; 5 s budget

Under the 5 s budget, the supplied-configuration path gives 255 definitive outcomes and 253 agreements among 258 inputs. VS126, VS127 and VS208 time out. Accuracy among definitive outcomes is 253/255; correct decisions over all inputs are 253/258. Figure 11 shows the two remaining definitive task-boundary differences, VS170 and VS300. The corrected VS215 and VS216 resonance regressions are concordant achiral cases and are therefore omitted from the figure. Chemical mirror identity compares the exact resonance family while preserving genuine bond-order differences. VS170 remains chiral because its literal `S-I` and `S#I` ligands are distinct; the ACS topology profile omits that distinction. VS300 is exactly achiral for the represented configured support: its mirror map fixes the central configured atom and

exchanges the two configured cage centers. The ACS topology-completion protocol adds global probes and therefore reaches its chiral label under a different support boundary. Neither difference is a search or certificate failure. Strict single-Lewis-state and ACS connectivity-only identities remain available as explicit profiles, and the strict library mode that audits undeclared supported loci remains separate from this supplied-configuration benchmark.

The new coupled-frame representation makes the VS300 distinction executable without rewriting the source tags. Its map-independent global certificate proves necessary chirality for sanitized VS300 topology but retains orientation as unspecified. Combining this proof with the unchanged exact source-declared procedure gives the separately named 257/258 protocol under a 30 s budget; VS170 remains the sole disagreement because chemical identity intentionally retains its bond-order distinction. This later result does not overwrite the frozen historical 256/258 artifact. Its stronger time budget must not be transferred to the five-second source-declared row. The comparator replays at pinned commit `2189f61` and current commit `2b99f62` each agree with the manual column on 254/258 rows. Their four disagreements are VS246, VS247, VS248 and VS299, separate from the published 258/258 column. A matched-inventory diagnostic transfers every upstream configured descriptor losslessly, including typed lone pairs, into the generic SynKit canonizer. Both algorithms then give the same mirror verdict on all four inputs. Removing the represented planar-bond descriptors changes both verdicts to achiral. An unrefined upstream stereo-map search also agrees with its fingerprint- assisted comparison. Thus the four-row discrepancy concerns the adapters' represented information, not evidence that canonicalization outperforms the upstream graph matcher. The ablation changes the configured object; it does not adjudicate the preferred chemical convention.

After stereo removal, the 258 records reduce to 214 unique constitutions. Completion assessment returns 125 necessarily chiral, 66 necessarily achiral, and 67

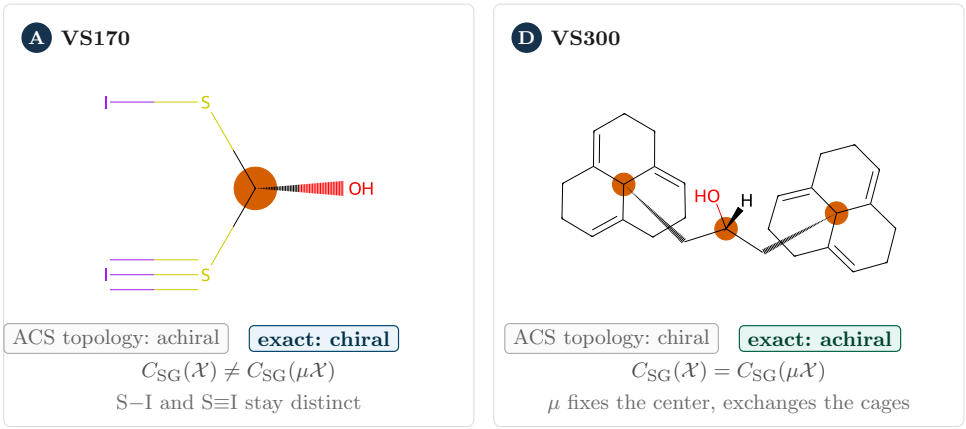


Fig. 11 The two remaining declared task-boundary differences in the 258-record source-declared-configuration audit. Orange circles mark atoms carrying explicit source `@/@@` configurations; wedge/hash bonds show the corresponding local orientation. In A, chemical identity preserves the genuine S−I/S≡I distinction and returns chiral, whereas the ACS connectivity-only profile returns achiral. In D, the exact mirror isomorphism fixes the central configured carbon and exchanges the two configured cages, so the represented local configurations compensate in a meso-like, mirror-fixed arrangement. This is not a general meso assignment: the historical source-declared protocol lacks the coupled relation. The separate global certificate encodes the topology-completion probes as one orientation-unspecified framework and proves the ACS chiral outcome without reinterpreting the three source tags.

configuration dependent, with no unsupported/incomplete row; all 258 rows are definitive. VS226 is settled by actual exhaustion of six symmetry-unique isomers. Automorphism-parity constraints prove that VS265 and VS266 have no achiral completion and construct an achiral completion of VS268. The erased manual label remains among the possible configurations for all 258 rows. This is a coverage statement, not recovery of information that was removed.

4 Discussion

4.1 Principal findings

Faithful reduction resolves simultaneous configurations and global symmetry within a declared identity profile. It preserves the distinction between E/Z and mirror relations and recognizes mirror-fixed *meso* structures without counting local labels. Formal

assignments are products of local orbits; global classes are their orbits under constitutional automorphisms. Corollary S13 justifies formal enumeration, and the retained raw encodings check its implementation. Independent partition and chemical studies support these declared contracts; perception, naming and performance results remain separate evidence.

4.2 Limitations

The method records assigned relative stereochemistry; it does not infer three-dimensional configuration, conformational populations, reaction selectivity, or configurational stability. Axis detection therefore does not establish that an atropisomer is isolable. Global/topological support is implemented for coupled tetrahedral frame certificates; additional framework geometries and automatic configured-orientation perception remain unsupported. The exact analyzer also fails closed above 256 support atoms or 64 coupled frames, and for frames distributed over disconnected components. Enhanced-stereo population semantics also remain unsupported.

The exact certificate consumes all ten configured families, but legacy value identity, rule matching, JSON/GML interchange, and the `RDKit` adapter still expose narrower class sets. Exact canonicalization materializes complete local orbits and uses verified automorphism pruning. Bounded correctness checks do not establish canonical-SMILES-scale performance; the controlled cost study explicitly retains resource-censored outcomes. Raw B is now complete, but raw C is represented only by the complete VS146 slice; the other eight raw C Cartesian products remain unexecuted and are not needed for the exact formal quotient. The CIP kernel implements Rules 1a, 1b, and 2, simple mancude averaging, Rule 3, and the reference-independent one-pair subset of Rules 4c/5; multi-unit Rule 4b and coordination-geometry projection remain fail-closed. Finally, ACS, RotA, and CIP provide different kinds of truth and cannot support one pooled accuracy statistic.

4.3 Future work

Priorities are sparse certificate construction and component decomposition, additional framework geometries and enhanced-stereo population semantics, and migration of extended carriers into legacy interchange surfaces. Experimental axis stability and handedness need separately sourced truth. Multi-unit CIP Rule 4b, hierarchical ring digraphs and coordinate-based orientation projections can improve naming coverage without changing identity.

5 Conclusions

The configured stereograph, rather than a collection of local names, is the appropriate identity object for molecular stereochemistry. The faithful auxiliary-graph expansion and complete canonical certificate decide global stereo-isomorphism exactly; the geometry-specific mirror action then makes whole-molecule chirality a theorem-backed consequence. Potential-carrier perception remains configuration neutral, authorized evidence is attached through a typed boundary, and incomplete chemistry fails closed. Finding a possible stereocenter is therefore only the first step and never, by itself, establishes molecular chirality.

List of abbreviations

ACS, American Chemical Society; API, application programming interface; CC BY, Creative Commons Attribution; CC BY-NC, Creative Commons Attribution Non-Commercial; CIP, Cahn–Ingold–Prelog; CPU, central processing unit; GML, Graph Modelling Language; ITS, imaginary transition state; JSON, JavaScript Object Notation; LLG, Lewis-labeled graph; RMSD, root-mean-square deviation; RSS, resident set size; SMILES, simplified molecular-input line-entry system; SP, square-planar; TBP, trigonal-bipyramidal; OH, octahedral.

Declarations

Ethics approval and consent to participate

Not applicable.

Consent for publication

Not applicable.

Availability of data and materials

The project is `SynKit`, with source code under the MIT License at its project home page [21]. General project documentation [22] and archived project releases [23] are publicly available; those releases do not identify the exact experimental revision described here. The ACS validation set is redistributed under CC BY-NC 4.0 and RotA under the MIT License. The original CIP publisher supporting archive, prepared SMILES and original coordinates are included under CC BY-NC 4.0 with attribution [6]; the license is separate from the software license. The later amended GitHub coordinate file is external and separately identified for historical replay. The selected MACE geometries and their attribution are included under CC BY 4.0 [20]; sourced negative inputs, retrieval responses, frozen annotations and production reports accompany the revision. The research checkout and package metadata report `SynKit 1.6.2`. Exact source commits and input checksums identify the revision; the package version alone does not identify the experimental code. The release material includes a Linux x86_64/Python 3.12.12 wheel-hash environment lock and the original conda bootstrap specification. The implementation is in Python; the tested environment uses RDKit 2026.03.1 and NetworkX 3.6.1, with a C++ compiler for the native comparator. The software has no additional restriction on use by non-academics; the separately licensed source datasets retain their stated terms. An exact review archive is provided locally

with a source-commit identifier and SHA-256 inventory; this review version has no public archive DOI.

Additional files

Additional file 1: `Additional_file_1.pdf` (PDF). *Supporting information: Exact canonicalization of configured molecular stereographs.* This file contains the technical algebra and proofs, representation and API contracts, benchmark protocols, chemical annotation details, resource measurements and reproduction instructions. It supplements the main article; machine-readable inputs, reports and software are retained in the separately identified research archive.

Competing interests

The authors' declaration of competing interests is pending.

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Authors' contributions

T.-L.P. conceived and developed the method, implemented the reference realization, designed and performed the validation, prepared the figures, and wrote the manuscript. P.F.S. and D.M. revised the manuscript, supervised the project, and secured funding.

Acknowledgements

`SynKit` builds on `RDKit` and `NetworkX`. The relative-stereo descriptor model was cross-validated against the MIT-licensed `StereoMolGraph` implementation; its notice and pinned conformance commit are distributed with the source.

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